

Neural interaction measures

Andrea Brovelli



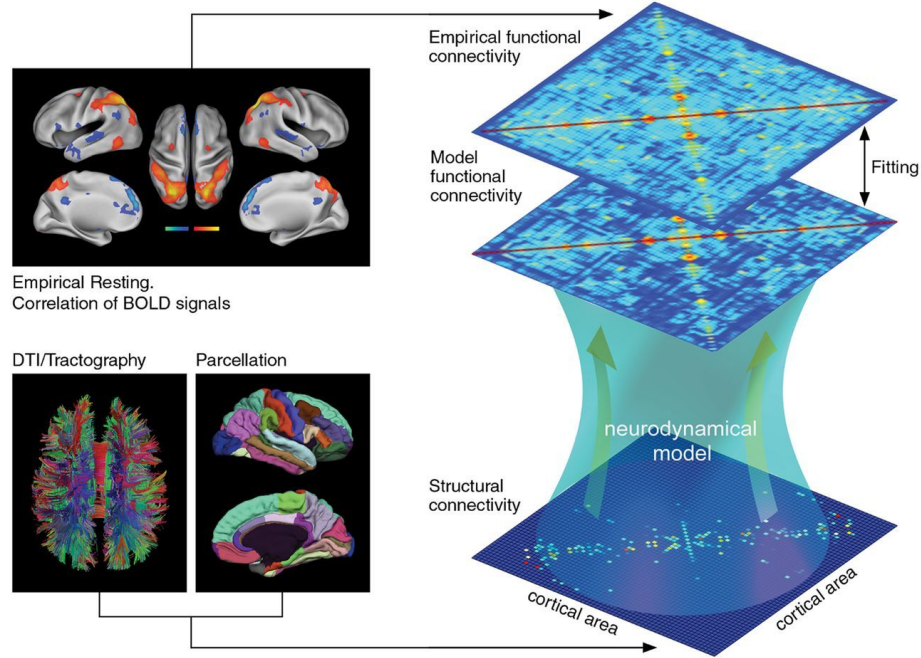
Institutions



Funding

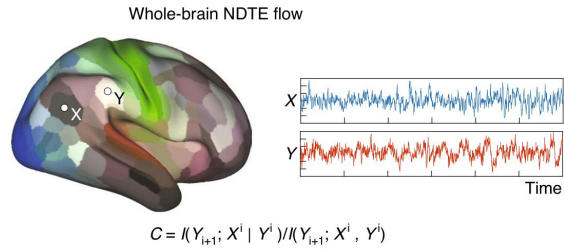


Brain models and neural interaction measures

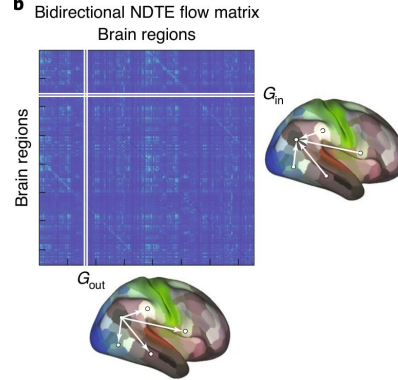


Brain models and neural interaction measures

a



b



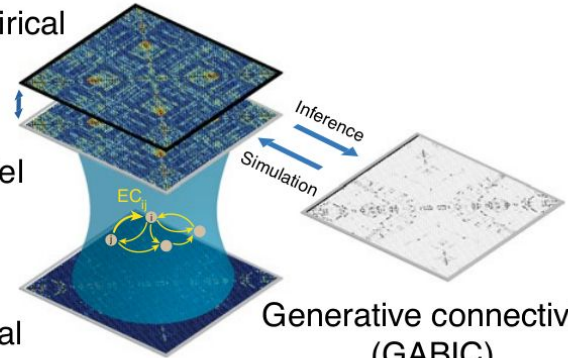
Creating whole-brain model

NDTE empirical

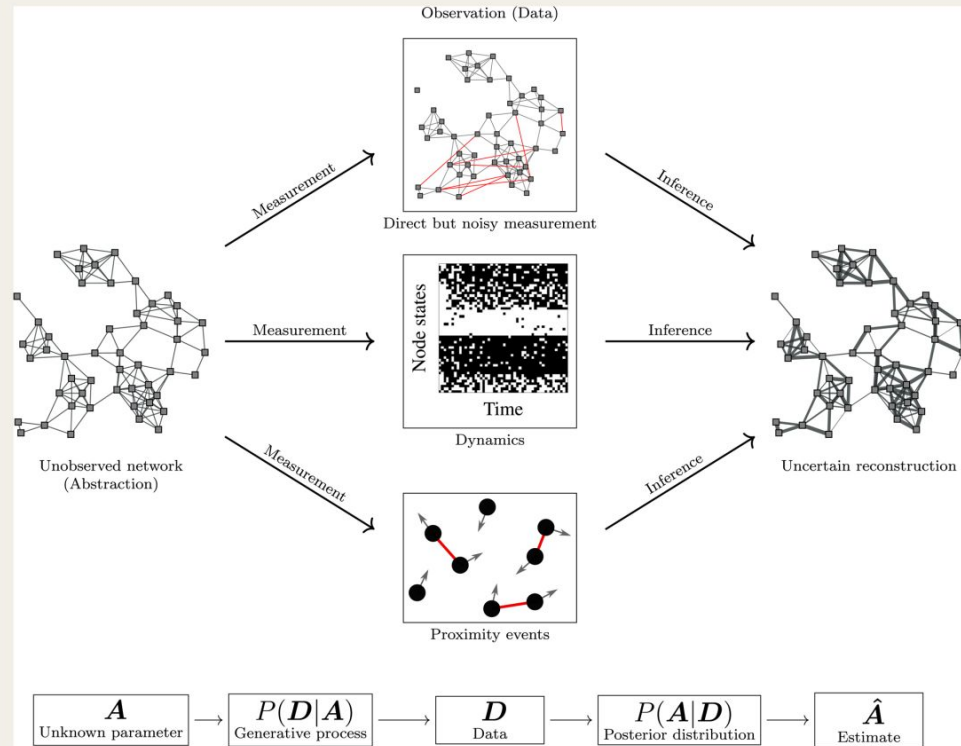
Fitting

NDTE model

SC empirical

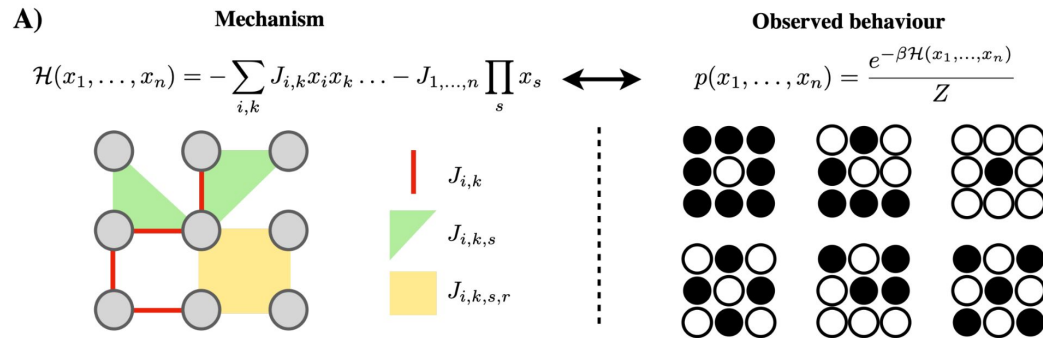


Linking data and theory in network science

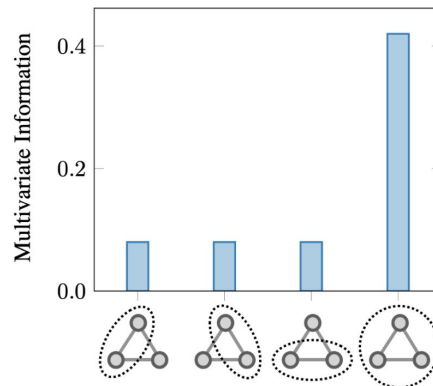
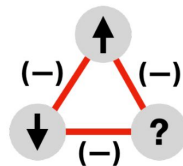


A network of interactions $\mathbf{A}$ that gives as a result some kind of observational data $\mathbf{D}$ should not in general be conflated with the data itself. Instead, we need to recognize that the data $\mathbf{D}$ is the result of measurement process $P(\mathbf{D}|\mathbf{A})$ that is conditioned on the unseen network, but is to some extent unavoidably decoupled from it. In order to estimate the underlying network, we need to perform an inferential step $P(\mathbf{A}|\mathbf{D})$, which needs to include our modeling assumptions about how the network and the data are generated. The resulting estimate $\hat{\mathbf{A}}$ will have an uncertainty that reflects the experimental design, accuracy of the measurements and overall feasibility of the particular reconstruction problem.

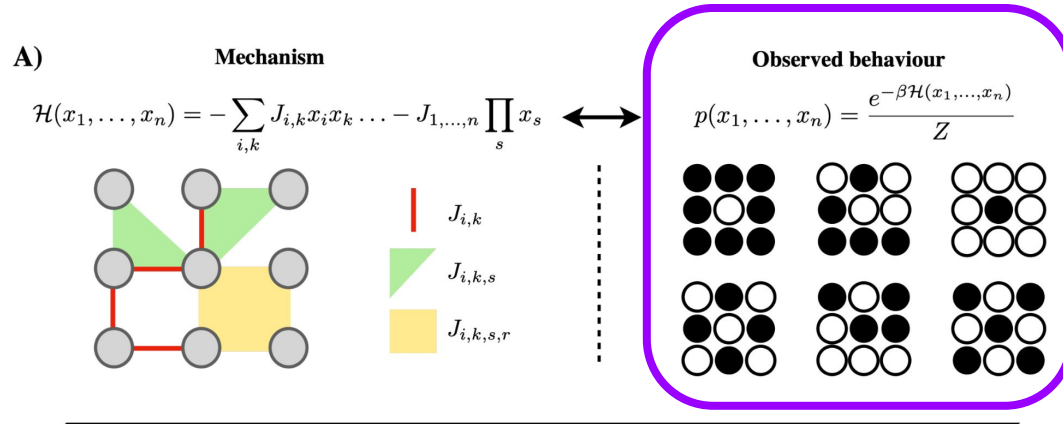
Linking “Mechanisms” and “Behaviours”



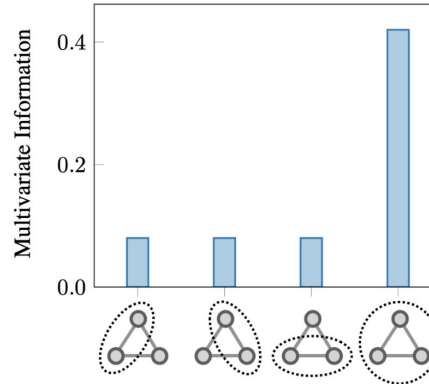
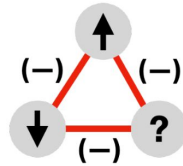
B) **Frustrated spin system**



Linking “Mechanisms” and “Behaviours”

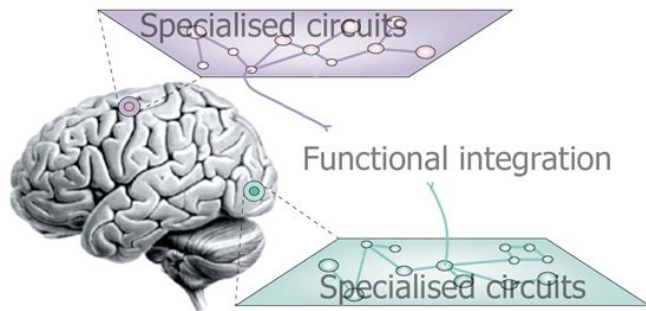


B) **Frustrated spin system**



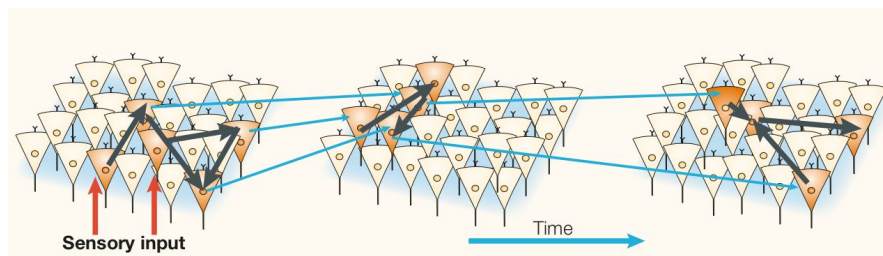
Collective behaviours and neural interactions across scales

The brainweb hypothesis



Varela et al (2001) *Nature Reviews Neurosci*

Cell assembly hypothesis (Hebb 1949)



Harris (2005) *Nature Reviews Neurosci*





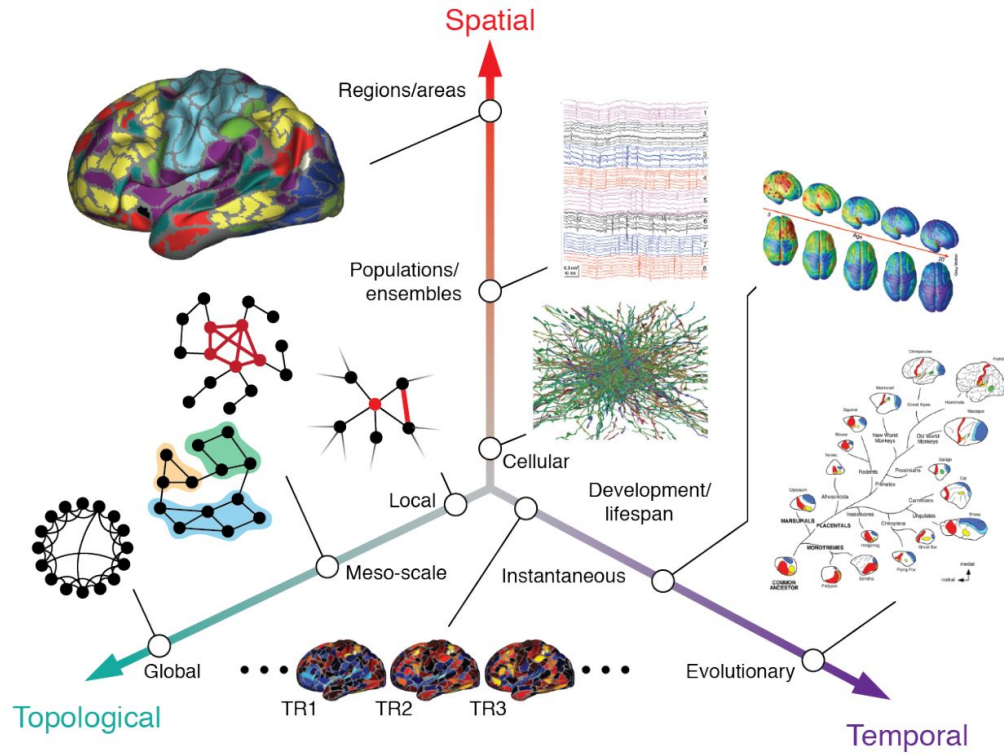
Q1. Interaction Rules

Q2. Functional Role

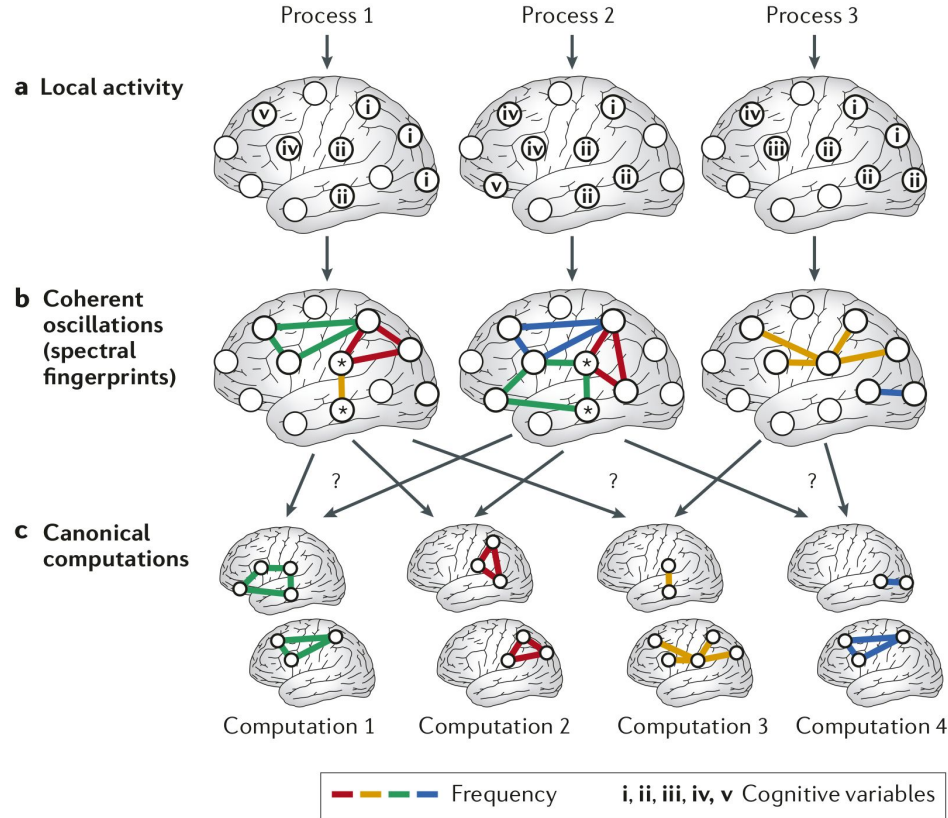
“Do as your neighbours”

Foraging, defense and travel

Neural interactions are complex

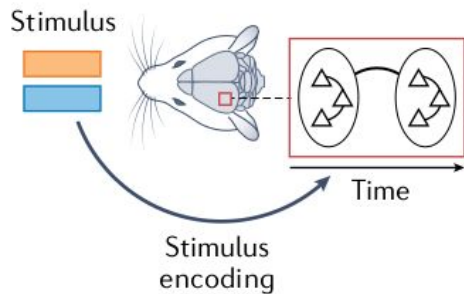


Neural oscillations and brain communication

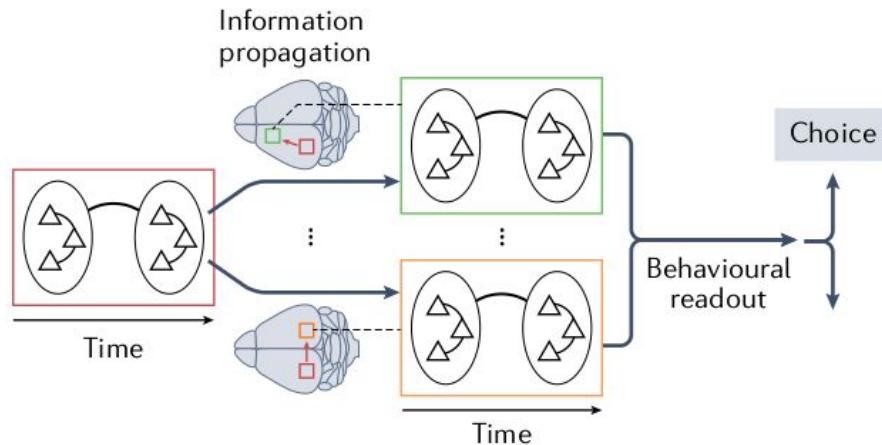


Why study neural interactions?

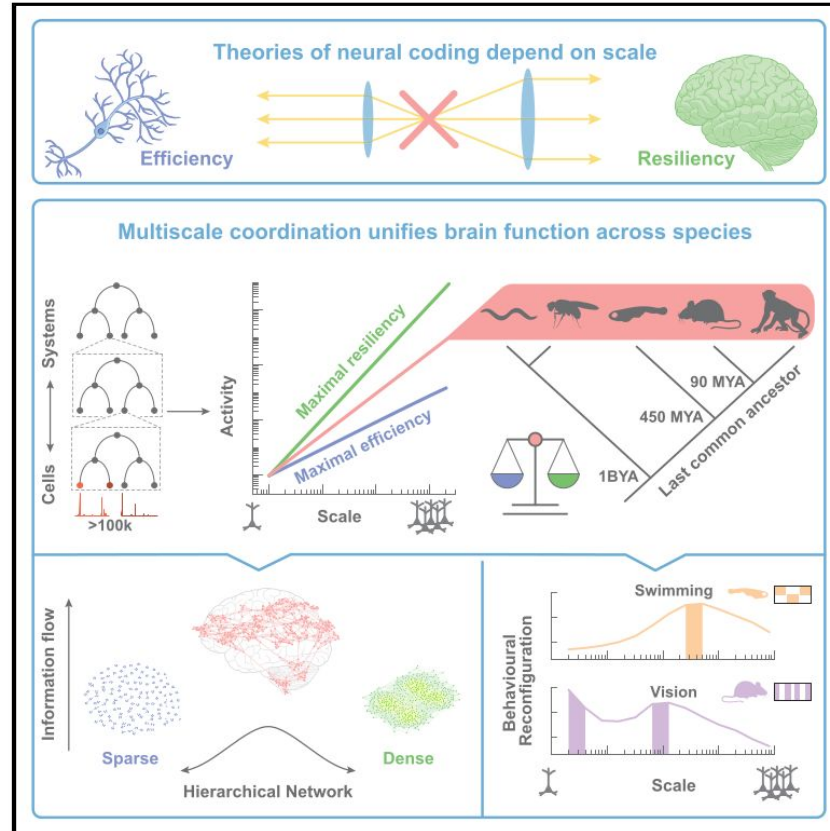
Distributed encoding of information relevant for cognition



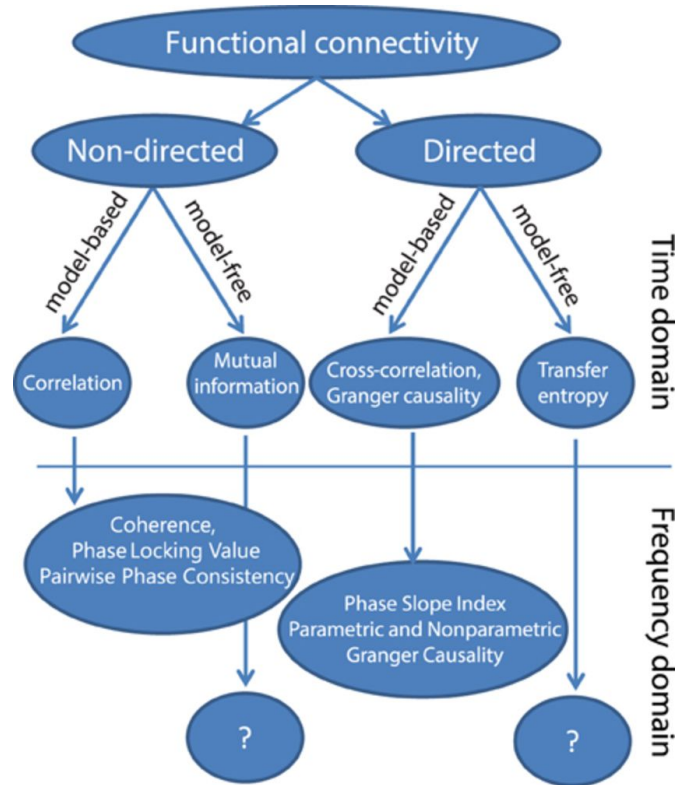
Broadcasting of information relevant for cognition



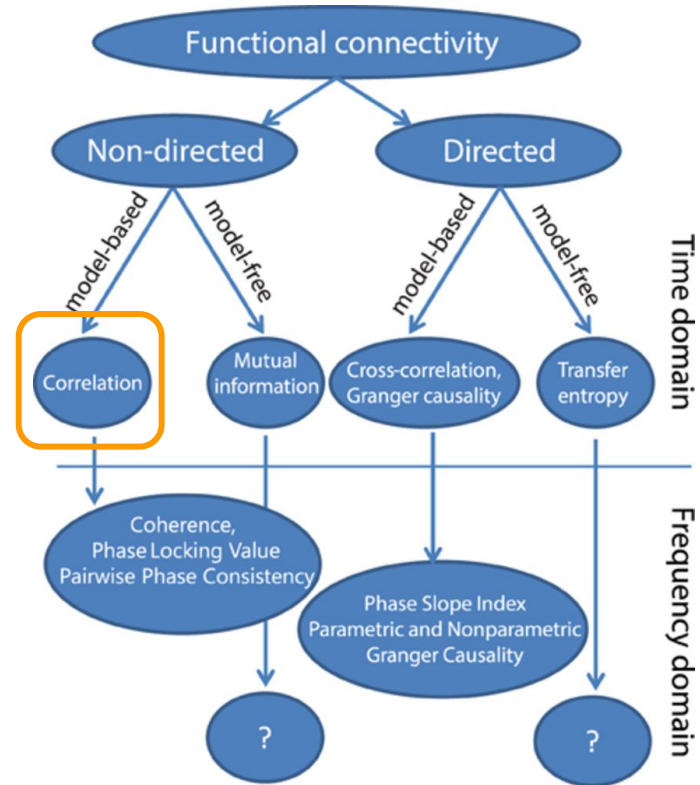
Efficient encoding across scales



Functional Connectivity measures

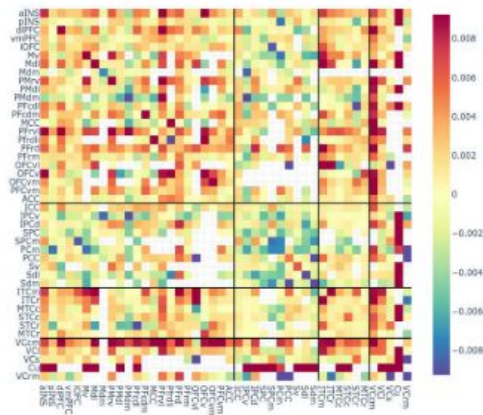
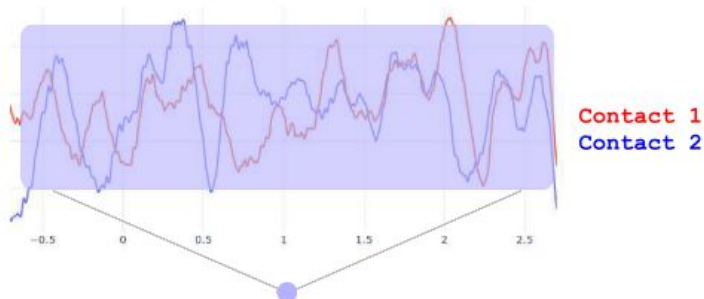


Functional Connectivity measures

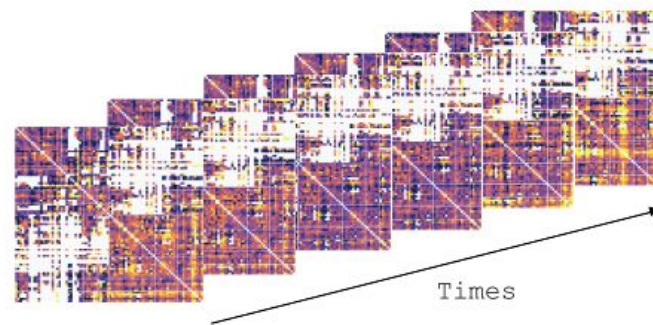
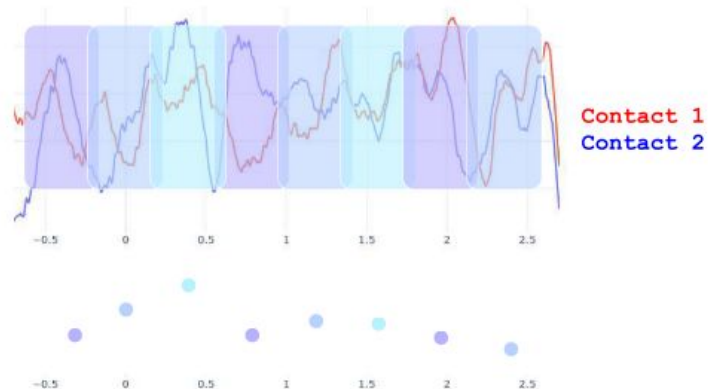


Functional Connectivity

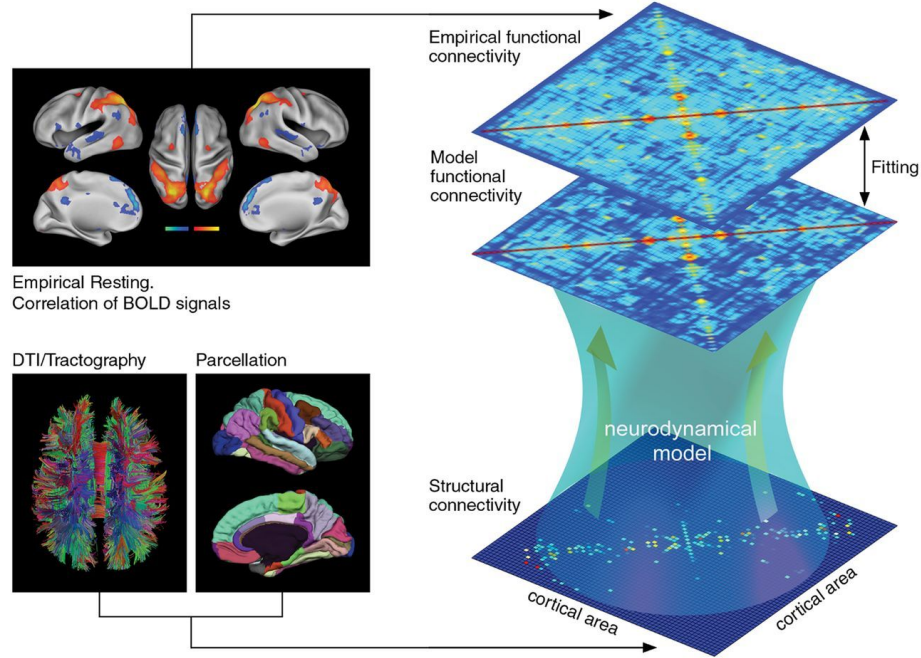
Static functional connectivity



Dynamic functional connectivity

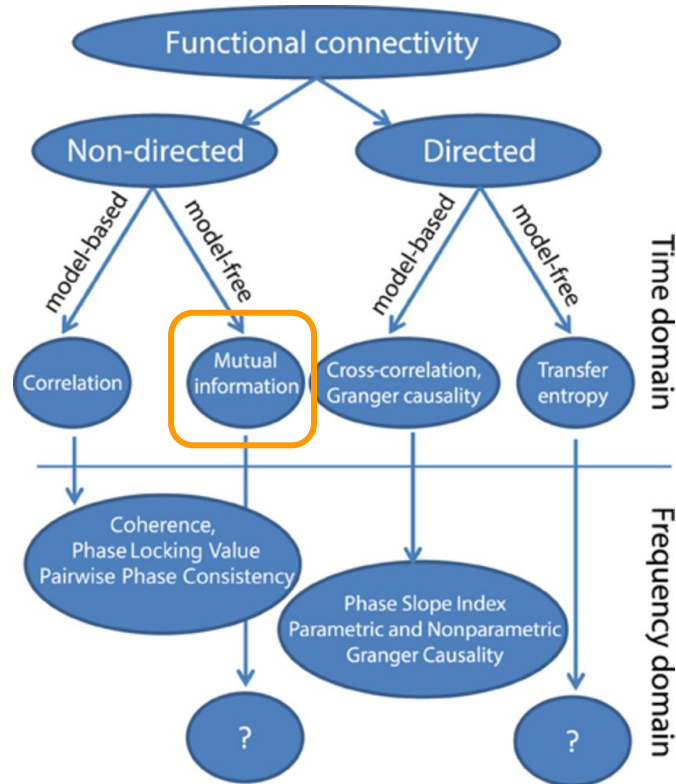


Brain models and neural interaction measures



Deco et al (2013) *JNeurosci*

Functional Connectivity measures



Information Theory



WIKIPEDIA
The Free Encyclopedia

Information theory (Shannon, 1948) is the scientific study of the quantification, storage, and communication of information

$$\textit{Surprisal} = -\log P(X)$$

Level of "surprise" of a particular outcome
Information content

$$H(X) = -\sum_{i=1}^N P(X_i) \log P(X_i)$$

Average uncertainty in X
Expected value of information carried by X

$$I(X; Y) = H(X) - H(X|Y)$$

Amount of information carried by Y about X,
and *vice versa*

$$I(X; Y|Z) = H(X; Z) - H(X|Z, Y)$$

Amount of information carried by Y about X
(and *vice versa*) given the value of a third
variable Z

Mutual Information

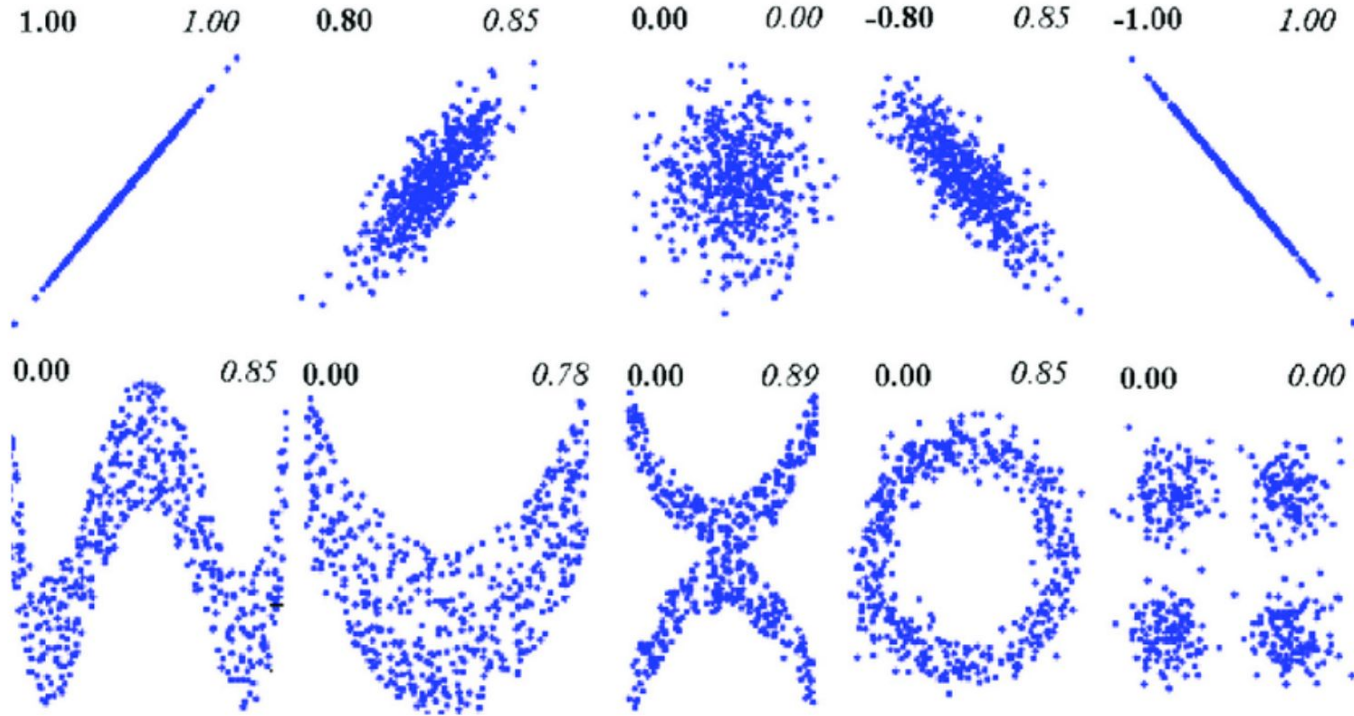
$H(X) = H(X|Y) \rightarrow$ **Zero** Mutual Dependence

$H(X) > H(X|Y) \rightarrow$ **Positive** Mutual Dependence

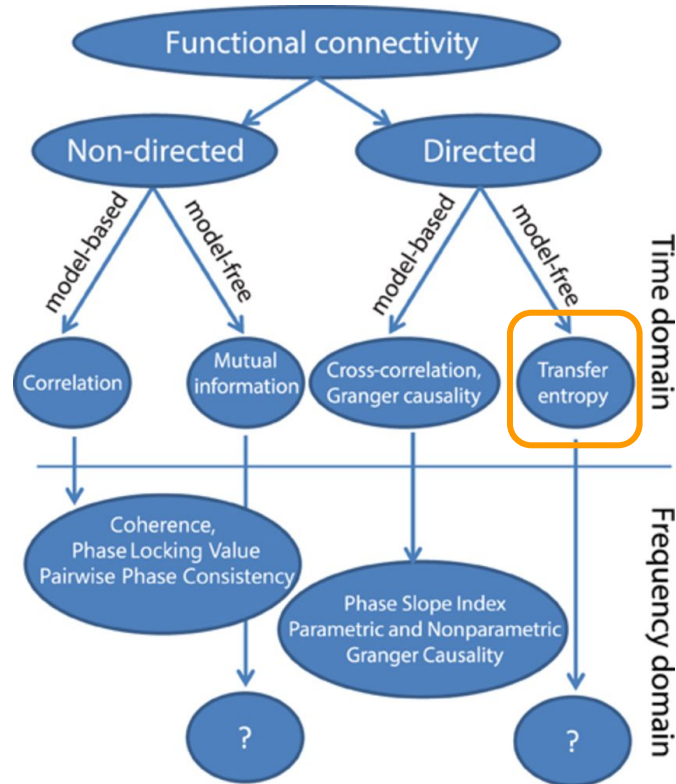
$$I(X; Y) = H(X) - H(X|Y)$$

Amount of mutual information between X and Y

Mutual Information vs Linear Correlation

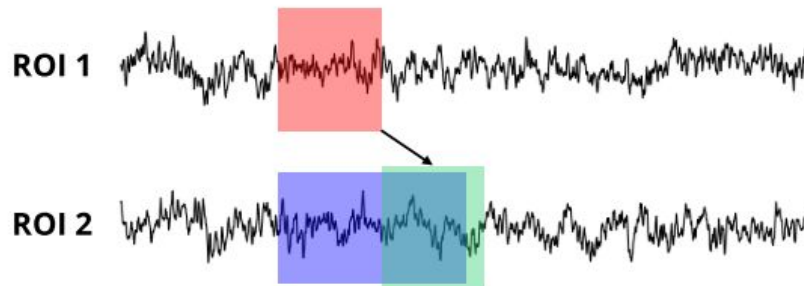


Functional Connectivity measures

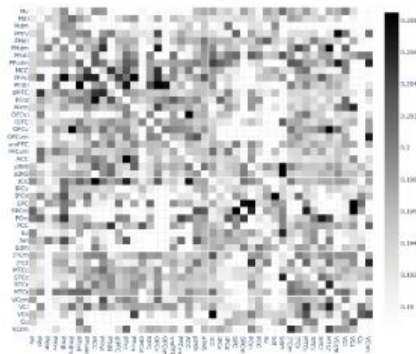


Transfer Entropy

$$I(\text{roi_1}_{\text{past}}; \text{roi_2}_{\text{present}} \mid \text{roi_2}_{\text{past}})$$



Static covGC

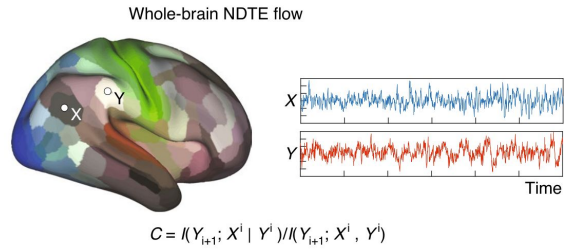


Dynamic covGC

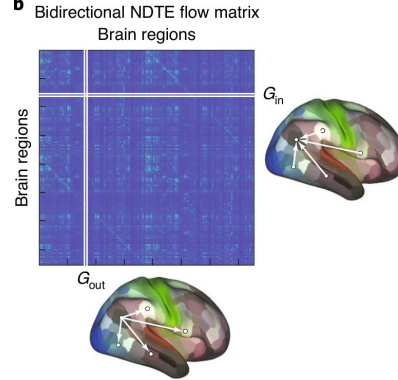


Brain models and neural interaction measures

a



b



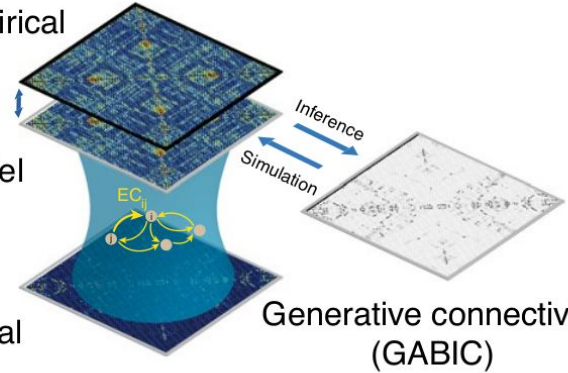
Creating whole-brain model

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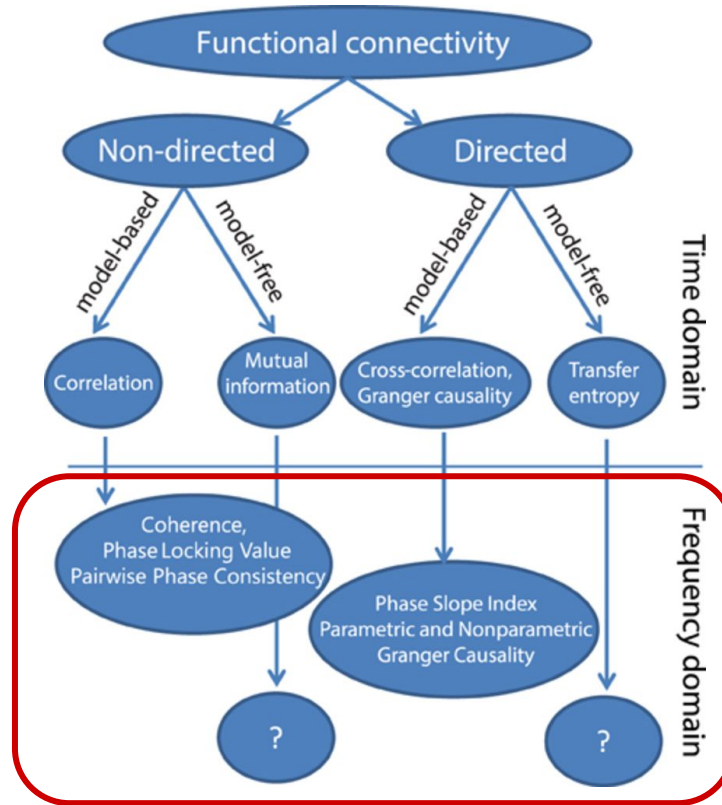
Fitting

NDTE model

SC empirical



Functional Connectivity measures



1. Information Decomposition

2. Higher-order interactions

**3. Feature-specific Information
Transfer**

1. Information Decomposition

2. Higher-order interactions

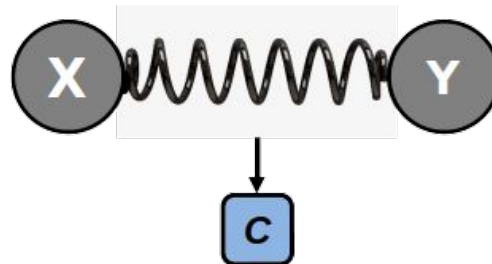
3. Feature-specific Information
Transfer

Task-related neural interactions

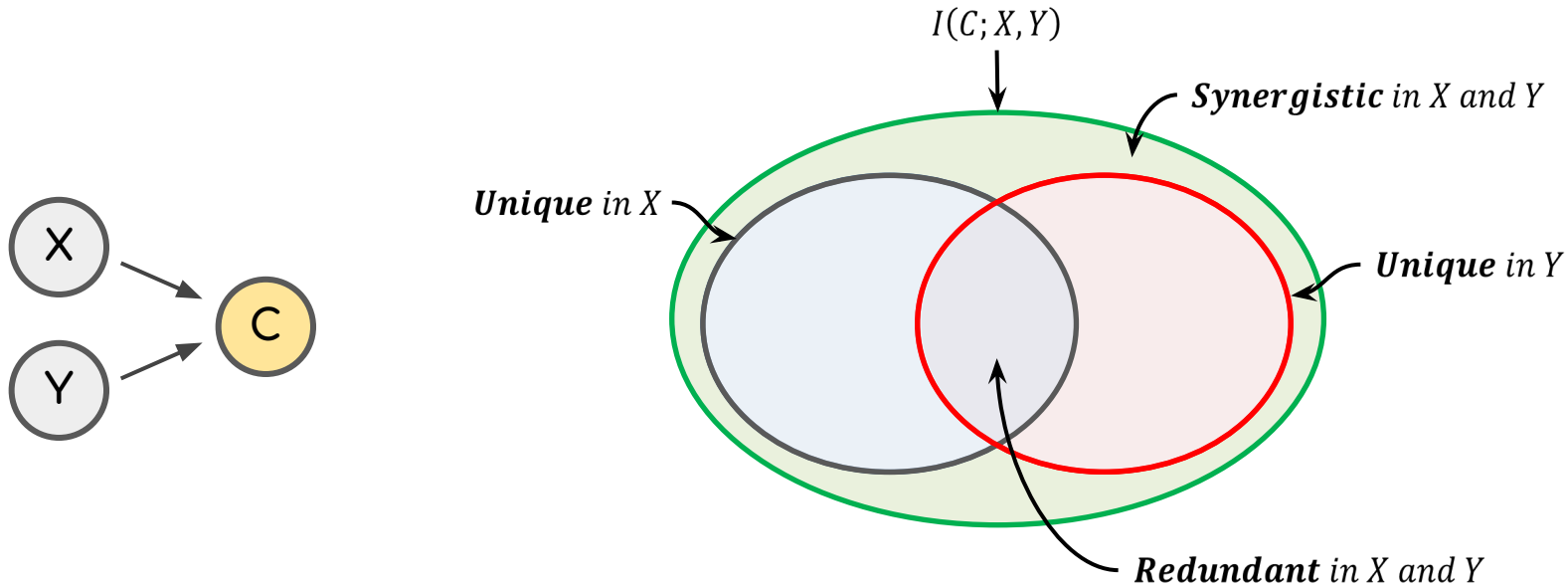
Amount of inter-areal dependence
(FC and FCD)



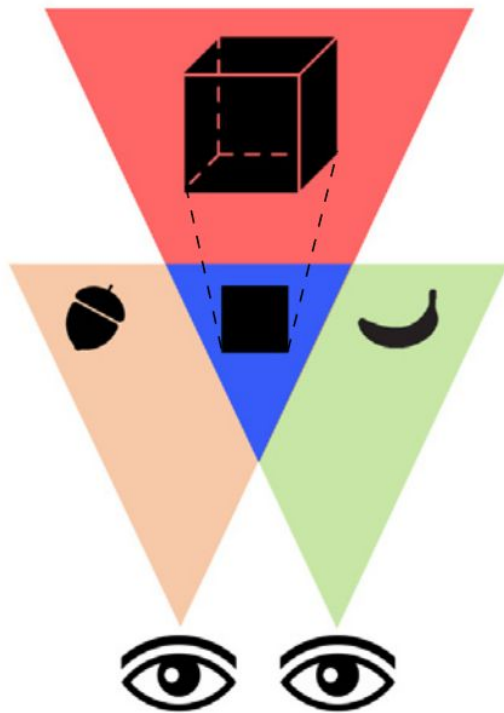
Amount and “Content”
of inter-areal dependence



Partial Information Decomposition



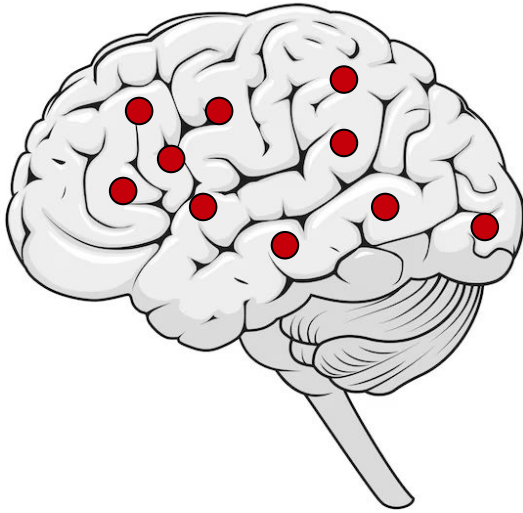
Information Decomposition



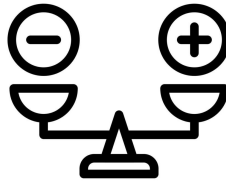
Luppi et al. (2024) *TINS*

Distributed encoding through brain interactions

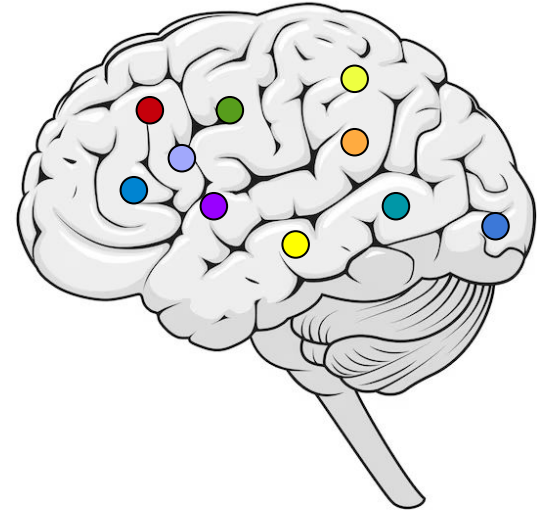
Redundant encoding



Pros: robust/resilient
Cons: inefficient



Synergistic encoding



Pros: efficient
Cons: fragile

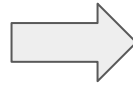
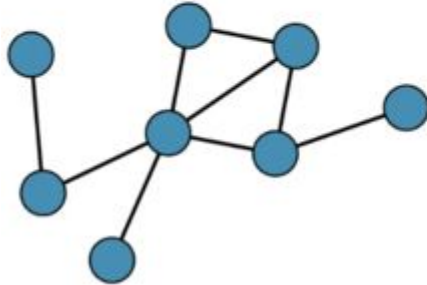
1. Information Decomposition

2. Higher-order interactions

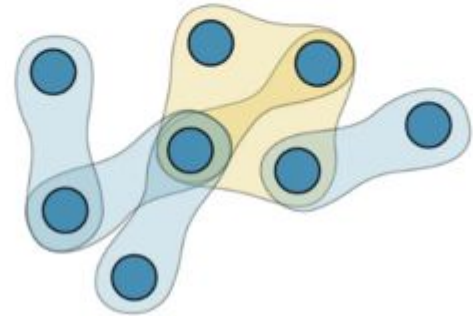
3. Feature-specific Information
Transfer

Graphs represents pairwise interactions

Pairwise interactions



Higher-order interactions



Network Science

nature
physics

PERSPECTIVE

<https://doi.org/10.1038/s41567-021-01371-4>

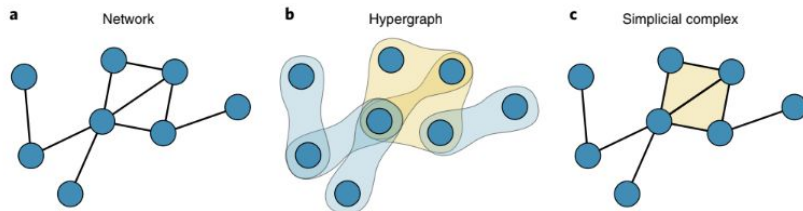


The physics of higher-order interactions in complex systems

Federico Battiston¹, Enrico Amico^{2,3}, Alain Barrat^{4,5}, Ginestra Bianconi^{6,7},
Guilherme Ferraz de Arruda⁸, Benedetta Franceschiello^{9,10}, Iacopo Iacopini¹, Sonia Kéfi^{11,12},
Vito Latora^{6,13,14,15}, Yamir Moreno^{8,15,16,17}, Micah M. Murray^{9,10,18}, Tiago P. Peixoto^{1,19},
Francesco Vaccarino²⁰ and Giovanni Petri^{8,21}

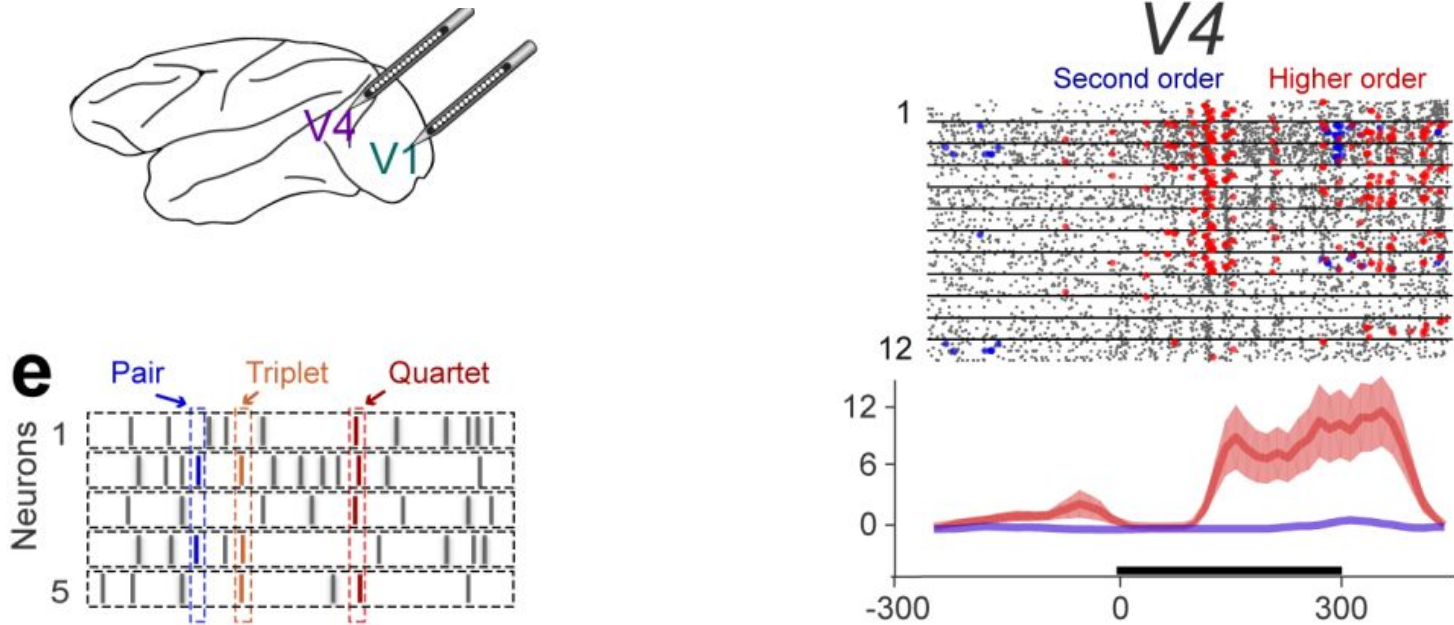
Complex networks have become the main paradigm for modelling the dynamics of interacting systems. However, networks are intrinsically limited to describing pairwise interactions, whereas real-world systems are often characterized by higher-order interactions involving groups of three or more units. Higher-order structures, such as hypergraphs and simplicial complexes, are therefore a better tool to map the real organization of many social, biological and man-made systems. Here, we highlight recent evidence of collective behaviours induced by higher-order interactions, and we outline three key challenges for the physics of higher-order systems.

How to represent HOIs

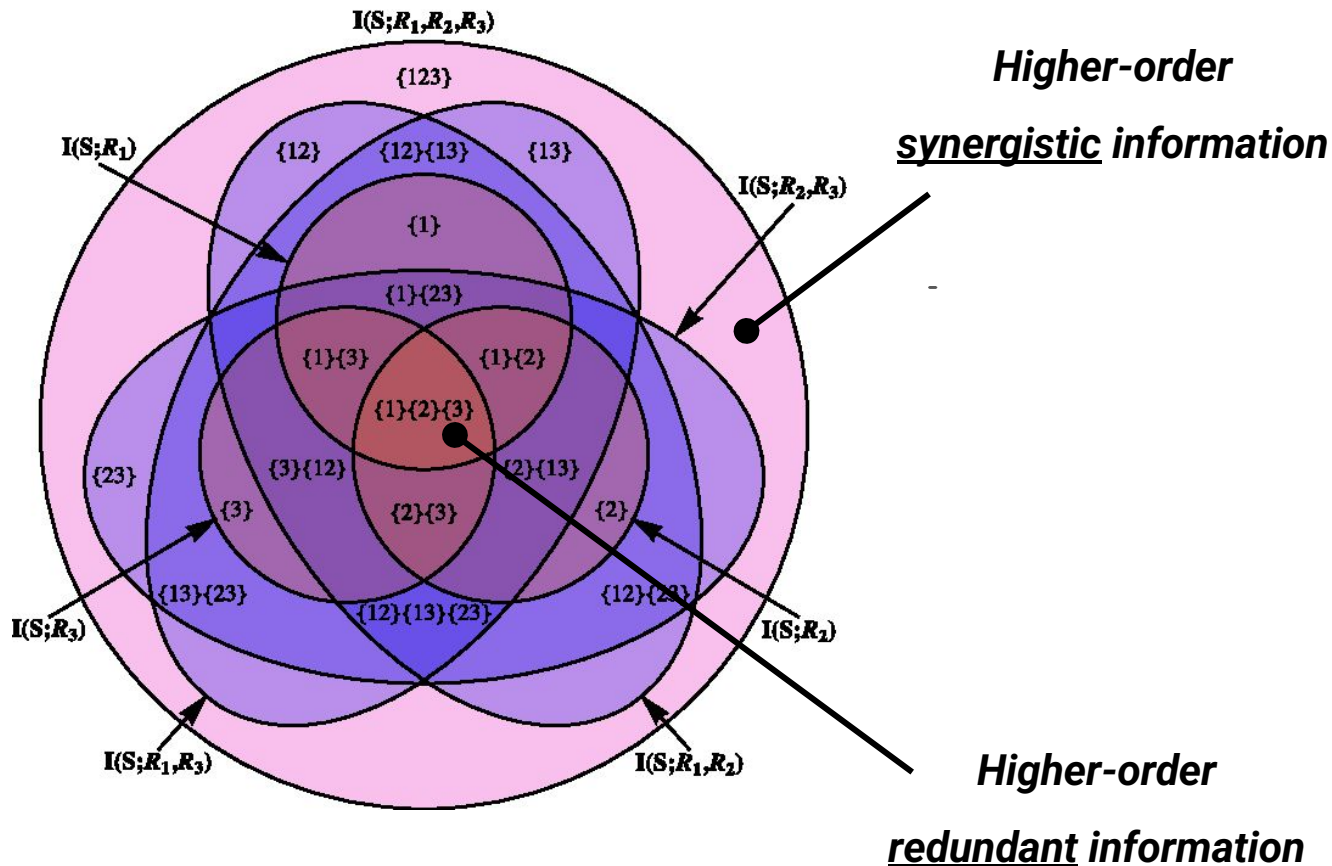
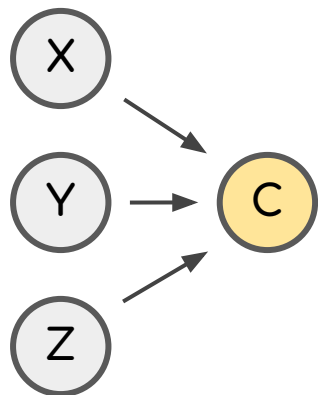


1.	Introduction.....
2.	Higher-order representations of networks.....
2.1.	Elementary representations of higher-order interactions.....
2.1.1.	Low- versus high-order representations
2.1.2.	Graph-based representations
2.1.3.	Explicit higher-order representations
2.2.	Relations and links between representations.....
3.	Measures.....
3.1.	Matrix representations of higher-order systems
3.1.1.	Incidence matrix.....
3.1.2.	Adjacency matrix.....
3.2.	Walks, paths and centrality measures
3.2.1.	Degree centralities
3.2.2.	Paths and path-based centralities
3.2.3.	Eigenvector centralities
3.3.	Triadic closure and clustering coefficient.....
3.4.	Simplicial homology
3.4.1.	Boundary operators and homology groups.....
3.4.2.	Evolving simplicial complexes.....
3.4.3.	Other measures of shape in simplicial complexes.....
3.5.	Higher-order Laplacian operators
3.5.1.	Hypergraph Laplacians
3.5.2.	Combinatorial Laplacians.....
4.	Models
4.1.	Equilibrium models
4.1.1.	Bipartite models
4.1.2.	Motifs models
4.1.3.	Stochastic set models
4.1.4.	Hypergraphs models
4.1.5.	Simplicial complexes models.....
4.2.	Out-of-equilibrium models
4.2.1.	Bipartite models
4.2.2.	Stochastic set models
4.2.3.	Hypergraphs models
4.2.4.	Simplicial complexes models.....
5.	Diffusion
5.1.	Higher-order diffusion.....
5.1.1.	Edge-flows.....

High-order collective behaviors in the brain



Higher-Order Information Decomposition

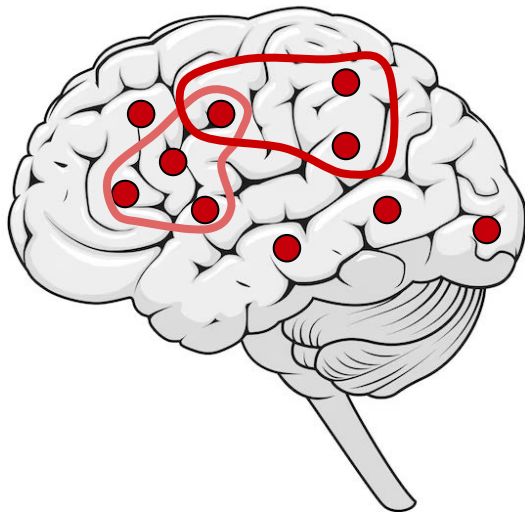


Dedekind number

$M(2) = 4$
 $M(3) = 18$
 $M(4) = 166$
 $M(5) = 7580$
 $M(6) = 7828354$
 $M(7) = 2414682040998$

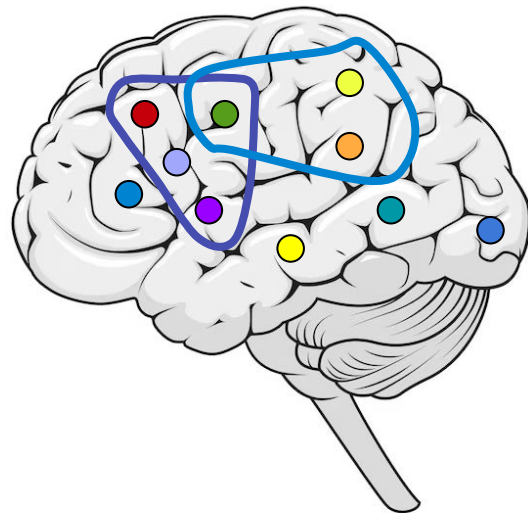
Encoding through higher-order brain interactions

Redundant encoding

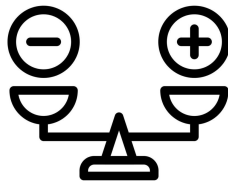


Pros: robust/resilient
Cons: inefficient

Synergistic encoding



Pros: efficient
Cons: fragile



Hebb, Hopfield, Singer, Abeles,
Bialek, Buzsaki, Aertsen, Varela,
Bressler, Deco, Breakspear, etc

Information Theory for collective behavior

PHYSICAL REVIEW E

covering statistical, nonlinear, biological, and soft matter physics

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Quantifying high-order interdependencies via multivariate extensions of the mutual information

Fernando E. Rosas, Pedro A. M. Mediano, Michael Gastpar, and Henrik J. Jensen
Phys. Rev. E **100**, 032305 – Published 13 September 2019



METHODS

published: 14 January 2021
doi: 10.3389/fphys.2020.595736



Quantifying Dynamical High-Order Interdependencies From the O-Information: An Application to Neural Spiking Dynamics

Sebastiano Stramaglia^{1,2*}, Tomas Scagliarini¹, Bryan C. Daniels³ and Daniele Marinazzo⁴

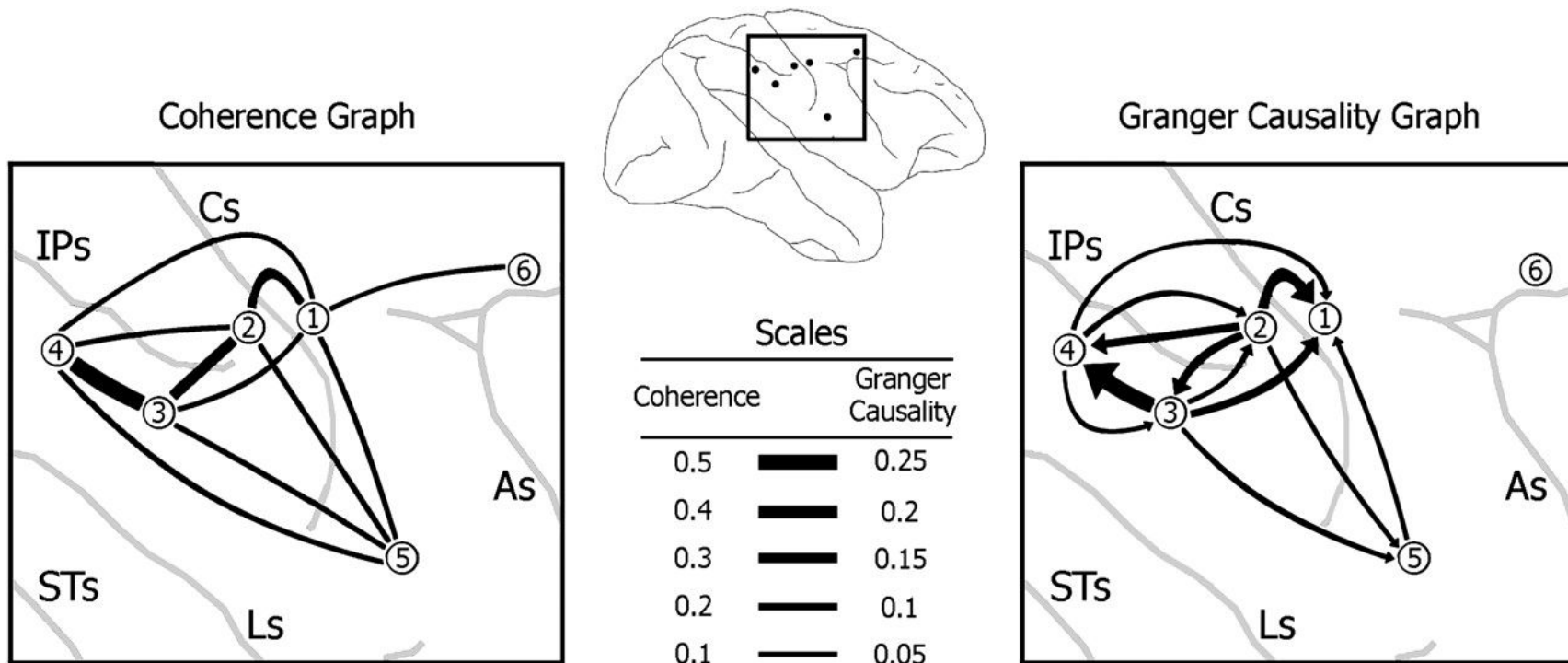
¹ Dipartimento Interateneo di Fisica, Università degli Studi Aldo Moro, Bari and INFN, Bari, Italy, ² Center of Innovative Technologies for Signal Detection and Processing (TIREs), Università degli Studi Aldo Moro, Bari, Italy, ³ Arizona State University and Santa Fe Institute Center for Biosocial Complex Systems, Arizona State University, Tempe, AZ, United States, ⁴ Department of Data Analysis, Ghent University, Ghent, Belgium

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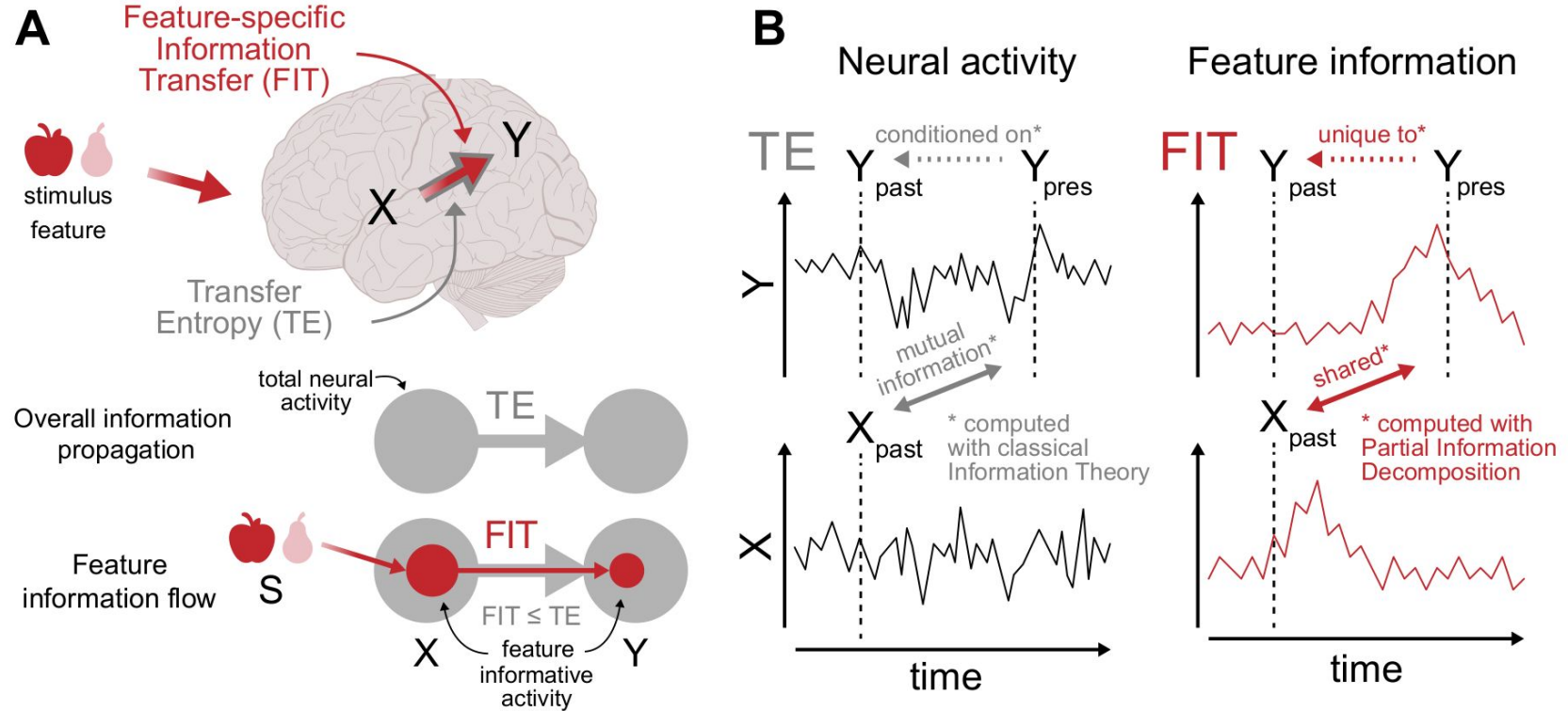
2. Higher-order interactions

**3. Feature-specific Information
Transfer**

Broadcasting of information

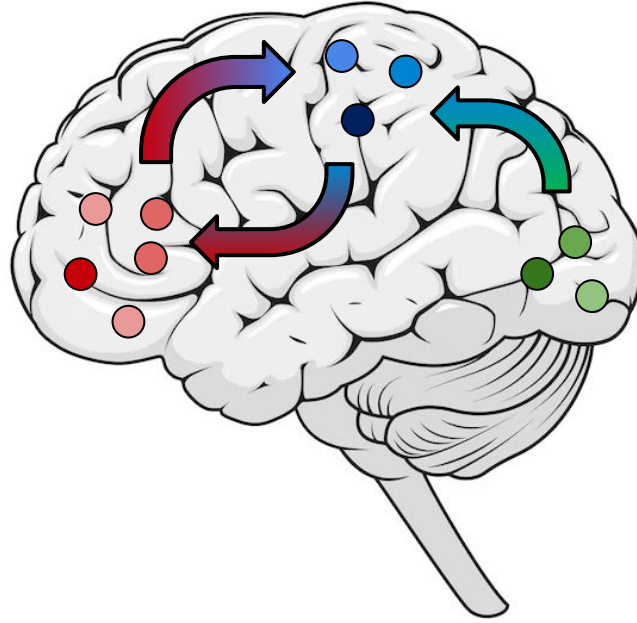


Broadcasting of information relevant for cognition



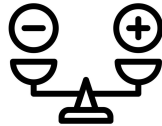
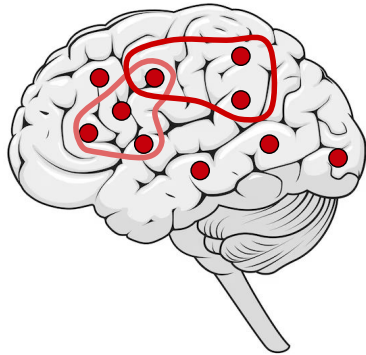
Broadcasting of information relevant for cognition

Task-specific information flow



Distributed higher-order encoding and broadcast of information relevant for cognitive functions

Redundant encoding

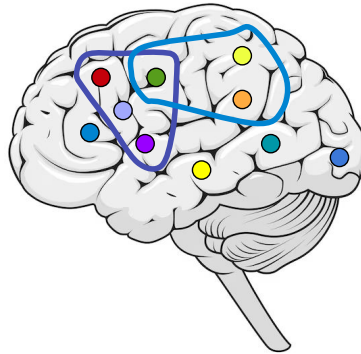


Pros: robust/resilient

Cons: inefficient

Hebb, Hopfield, Singer, Abeles, Bialek,
Buzsaki, Aertsen, Varela, Bressler,
Deco, Breakspear, etc

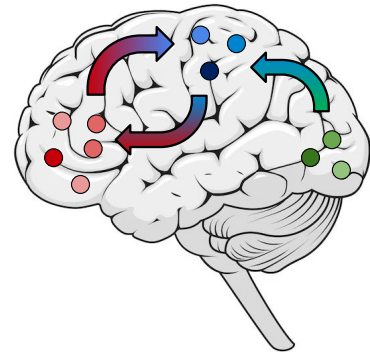
Synergistic encoding



Pros: efficient

Cons: fragile

Information Flow



Fries, Bressler, Vinck, Panzeri

1. Neural interactions in the human frontal cortex dissociate reward and punishment learning

Combrisson E, Basanisi R, Gueguen MCM, Rheims S, Kahane P, Bastin P, Brovelli A
eLife (2024)

2. Higher-order and synergistic functional interactions encode information gain in goal-directed learning

Combrisson E*, Basanisi R*, Neri M, Auzias G, Petri G, Marinazzo D, Panzeri S, Brovelli A
Nat Commun (2025)

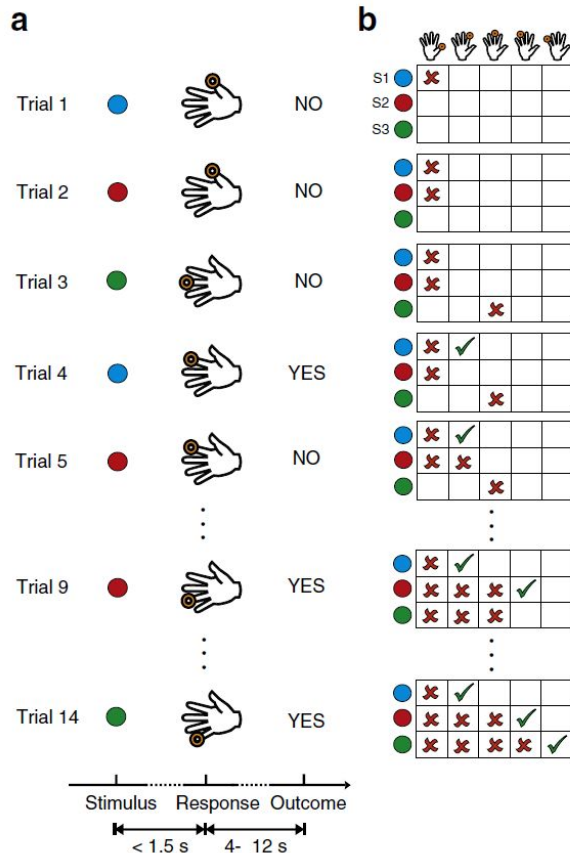
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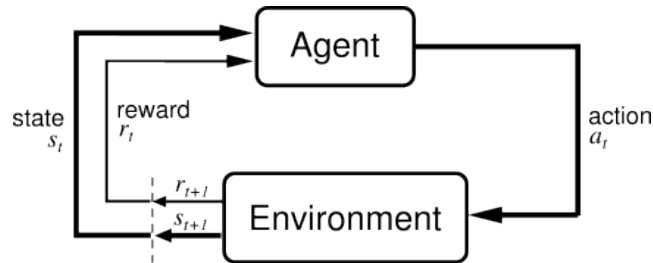
Goal-directed information seeking task



*Deterministic causal
relations between
stimuli, actions and
outcomes*

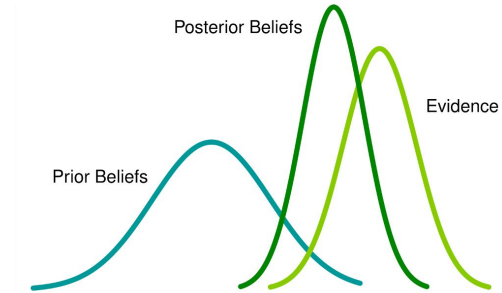
Learning theories

Reinforcement Learning



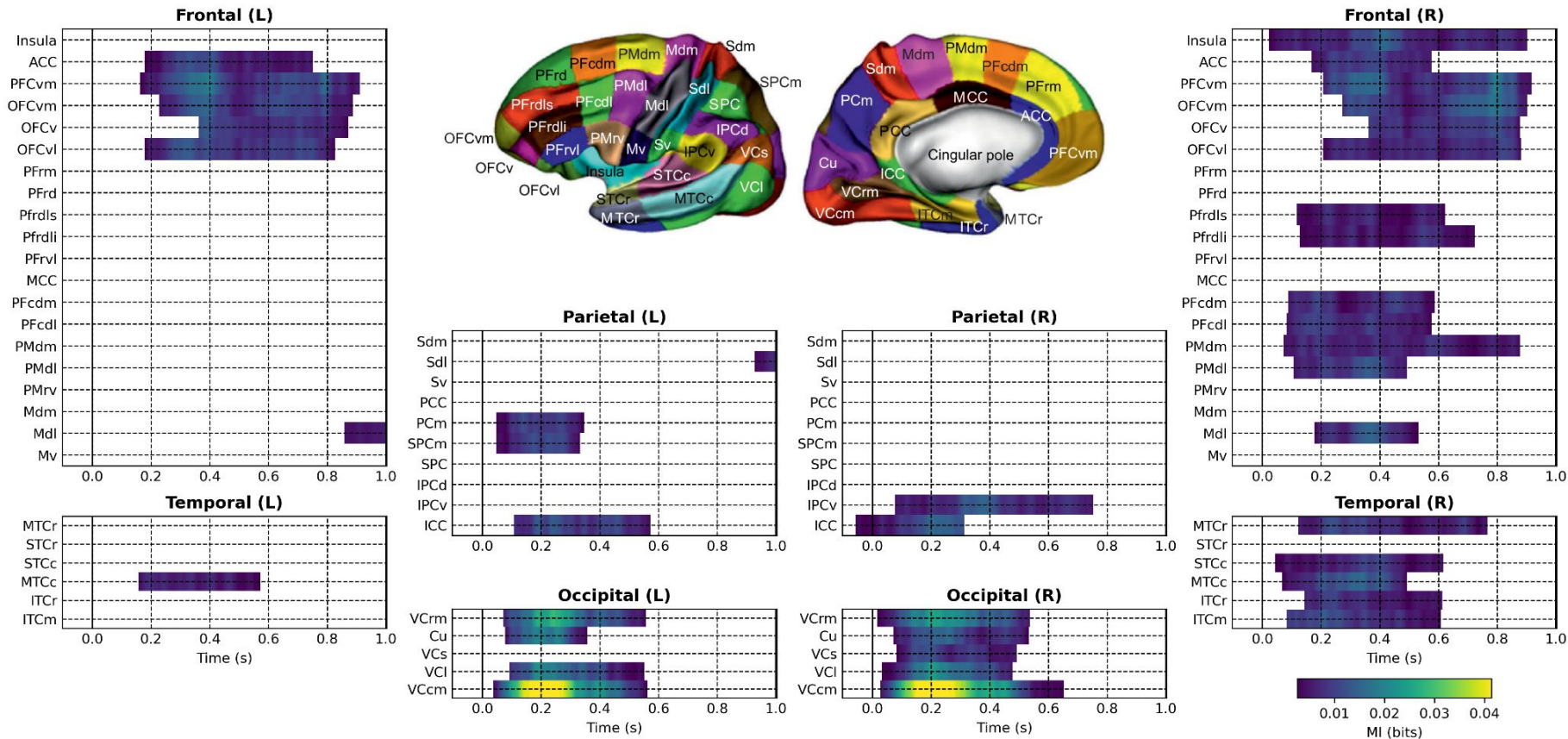
Reward Prediction Errors

Bayesian Inference

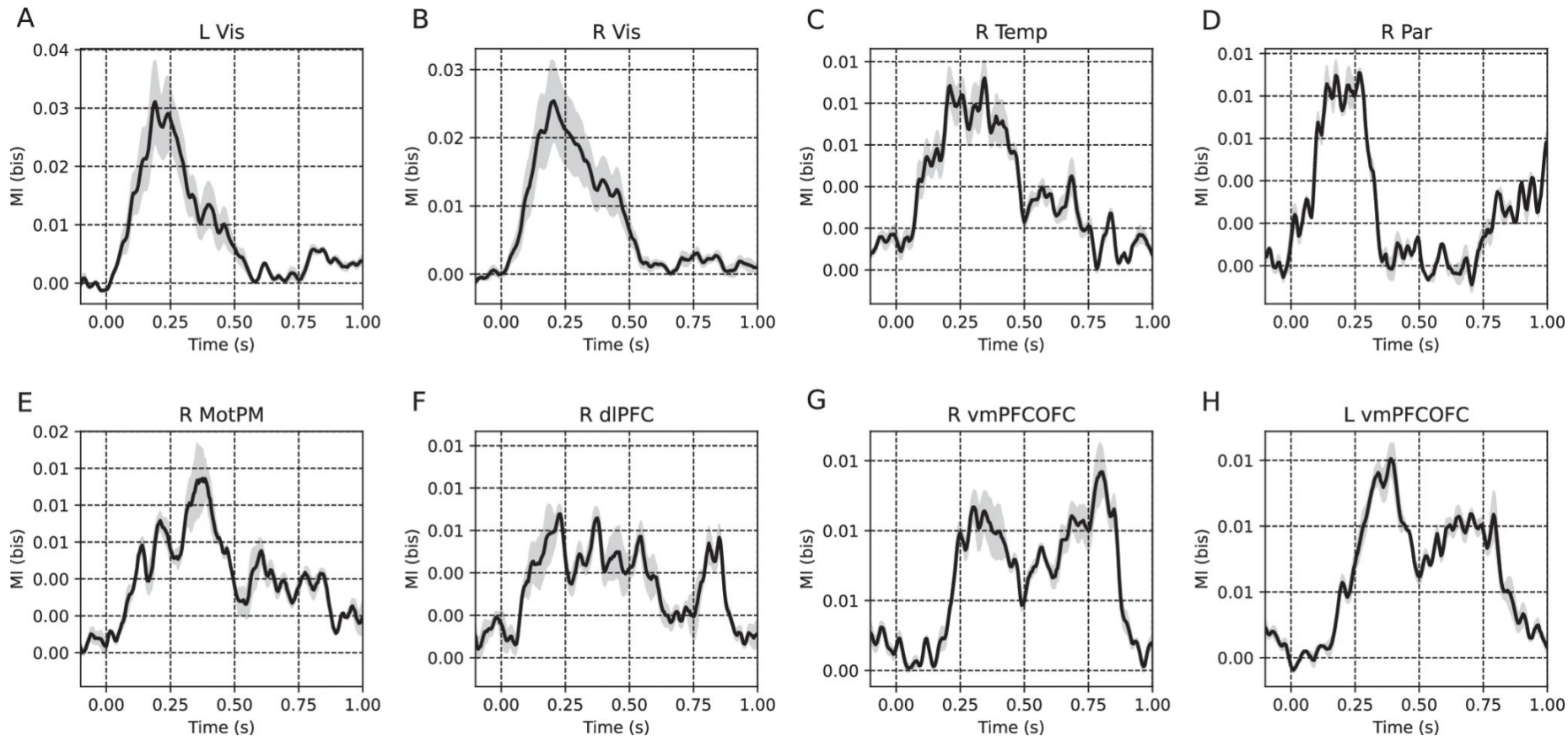


Information Gain

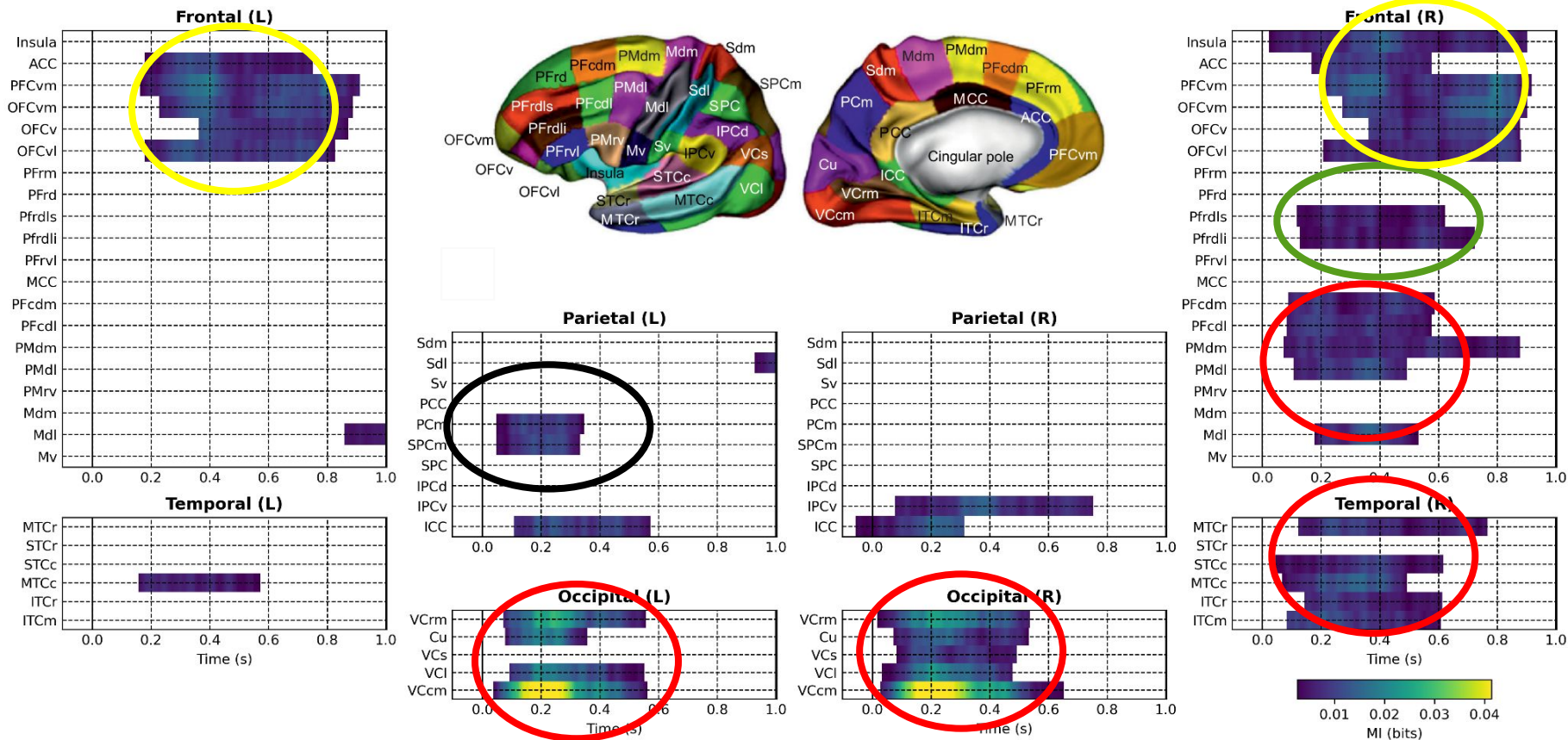
Local encoding of Information Gain



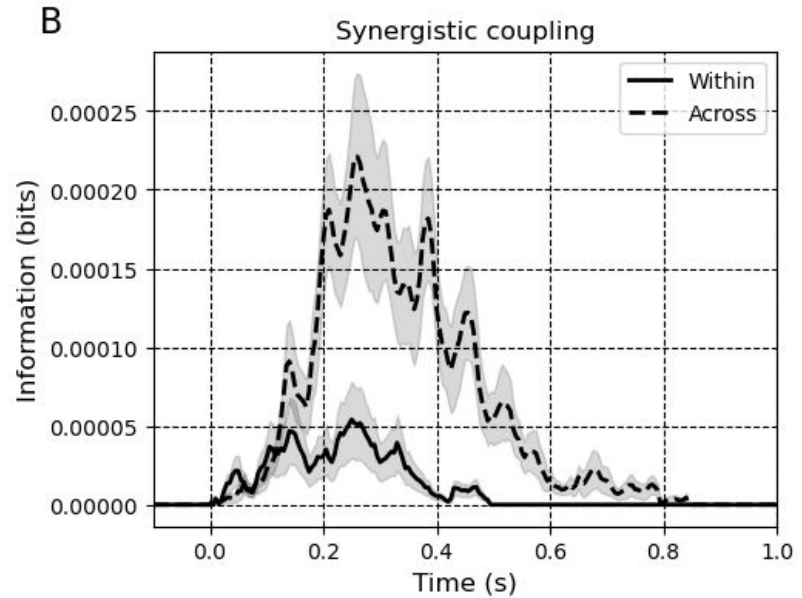
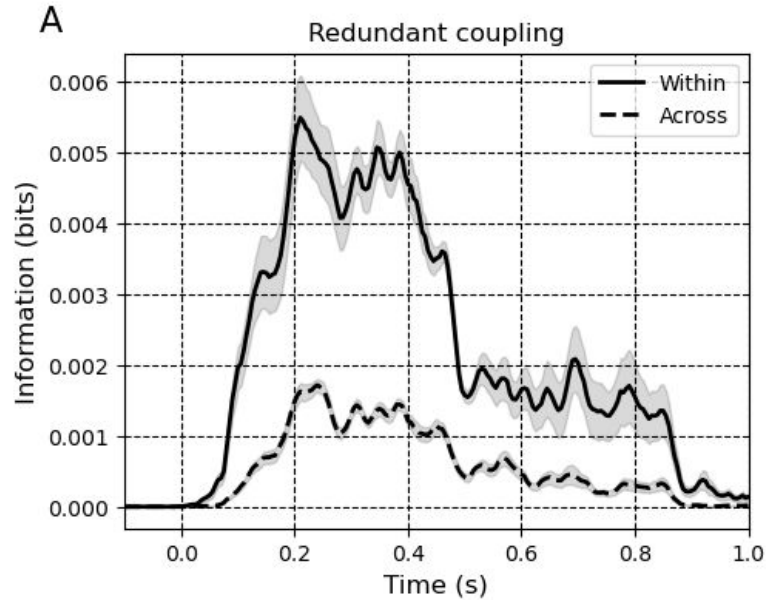
Local encoding of Information Gain



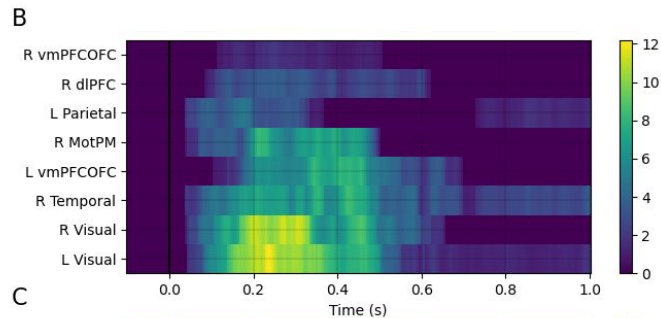
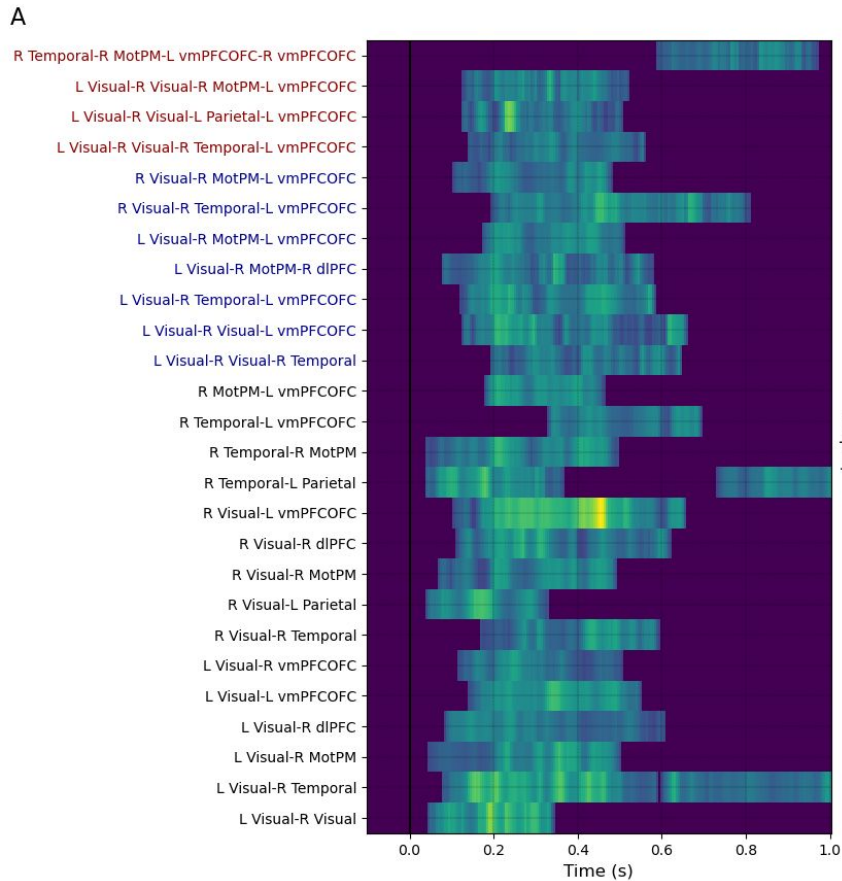
Integration versus Segregation?



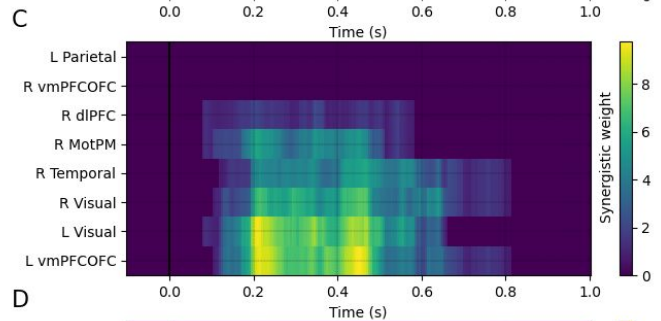
Integration *versus* Segregation?



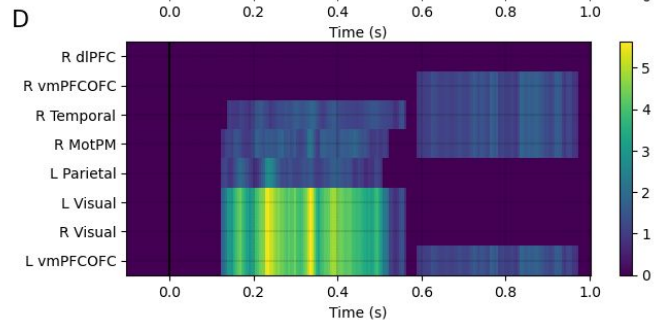
Higher-order and synergistic encoding



Pairwise



Triplets

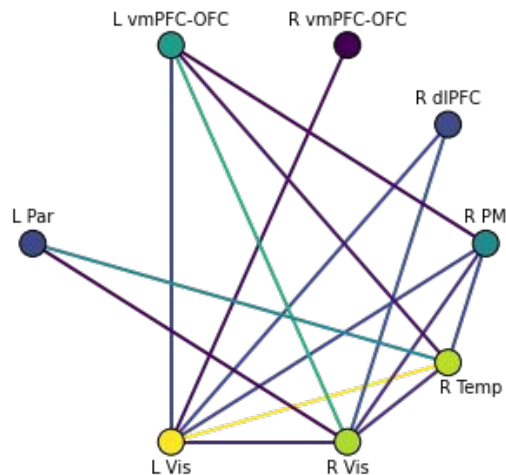


Quadruplets

Higher-order overlap and synergistic encoding

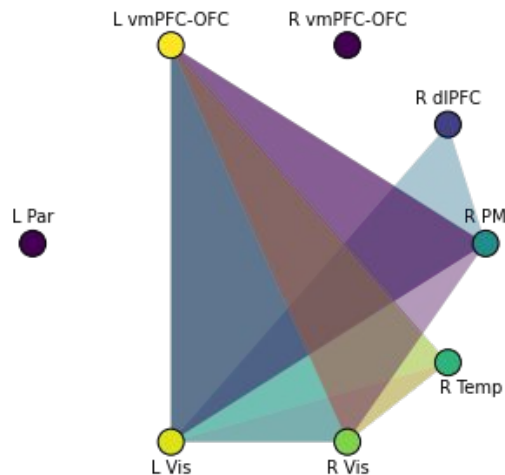
Pairwise

A



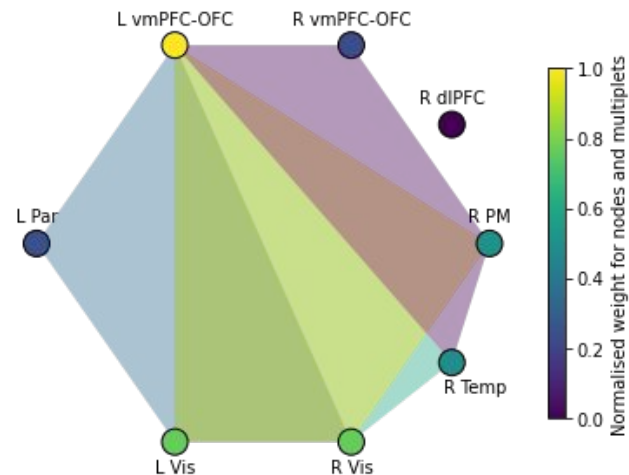
Triplets

B

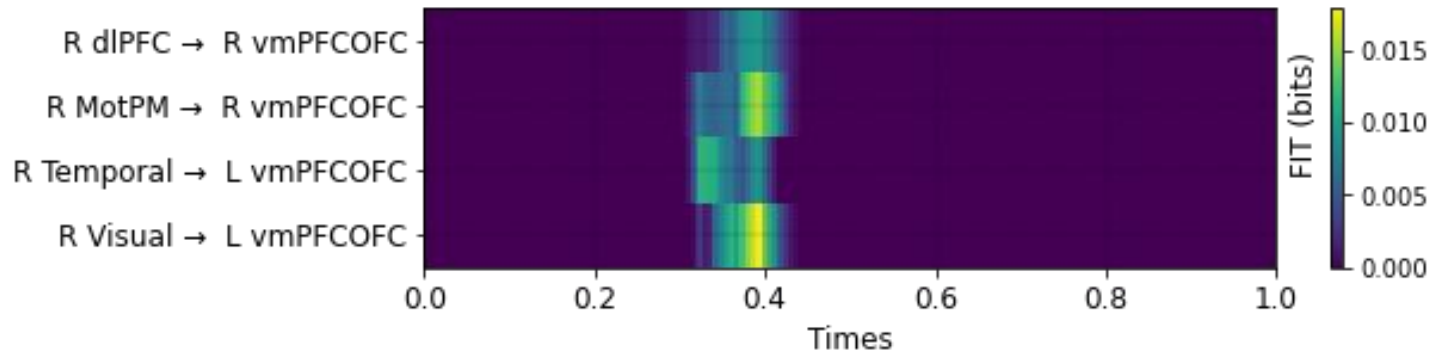


Quadruplets

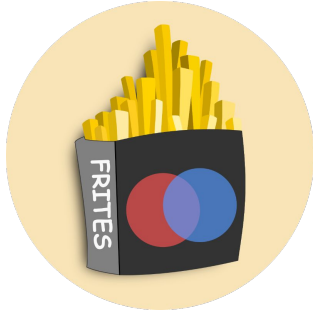
C



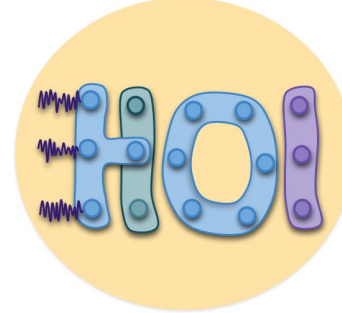
Broadcasting of information along HOI synergistic interactions



Try it at home!



FRamework for
Information
Theoretical analysis of
Electrophysiological data and
Statistics



Higher
Orders
Interactions



<https://github.com/brainets/frites>



<https://brainets.github.io/frites/>

Combrisson et al (2022) *JOSS*



<https://github.com/brainets/hoi>



<https://brainets.github.io/hoi/>

Neri et al (2024) *JOSS*



EBRAINS



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Group-level inference of information-based measures for the analyses of cognitive brain networks from neurophysiological data



Etienne Combrisson^{a,*}, Michele Allegra^{a,d,e}, Ruggero Basanisi^a, Robin A.A. Ince^b,
Bruno L. Giordano^a, Julien Bastin^c, Andrea Brovelli^{a,*}

^a Aix Marseille Univ, CNRS, INT, Institut de Neurosciences de la Timone, Marseille, France

^b School of Psychology and Neuroscience, University of Glasgow, Glasgow, UK

^c Univ. Grenoble Alpes, Inserm, U1216, Grenoble Institut Neurosciences, 38000 Grenoble, France

^d Dipartimento di Fisica e Astronomia "Galileo Galilei", Università di Padova, via Marzolo 8, Padova 35131, Italy

^e Padua Neuroscience Center, Università di Padova, via Orus 2, Padova 35131, Italy

Frites: A Python package for functional connectivity analysis and group-level statistics of neurophysiological data

Etienne Combrisson ¹, Ruggero Basanisi ¹, Vinicius Lima Cordeiro ^{1,2}, Robin A. A Ince ³, and Andrea Brovelli ¹

¹ Institut de Neurosciences de la Timone, Aix Marseille Université, UMR 7289 CNRS, 13005, Marseille, France ² Institut de Neurosciences des Systèmes, Aix-Marseille Université, UMR 1106 Inserm, 13005, Marseille, France ³ Institute of Neuroscience and Psychology, University of Glasgow, Glasgow, UK

DOI: [10.21105/joss.03842](https://doi.org/10.21105/joss.03842)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Marie E. Rognes](#)  

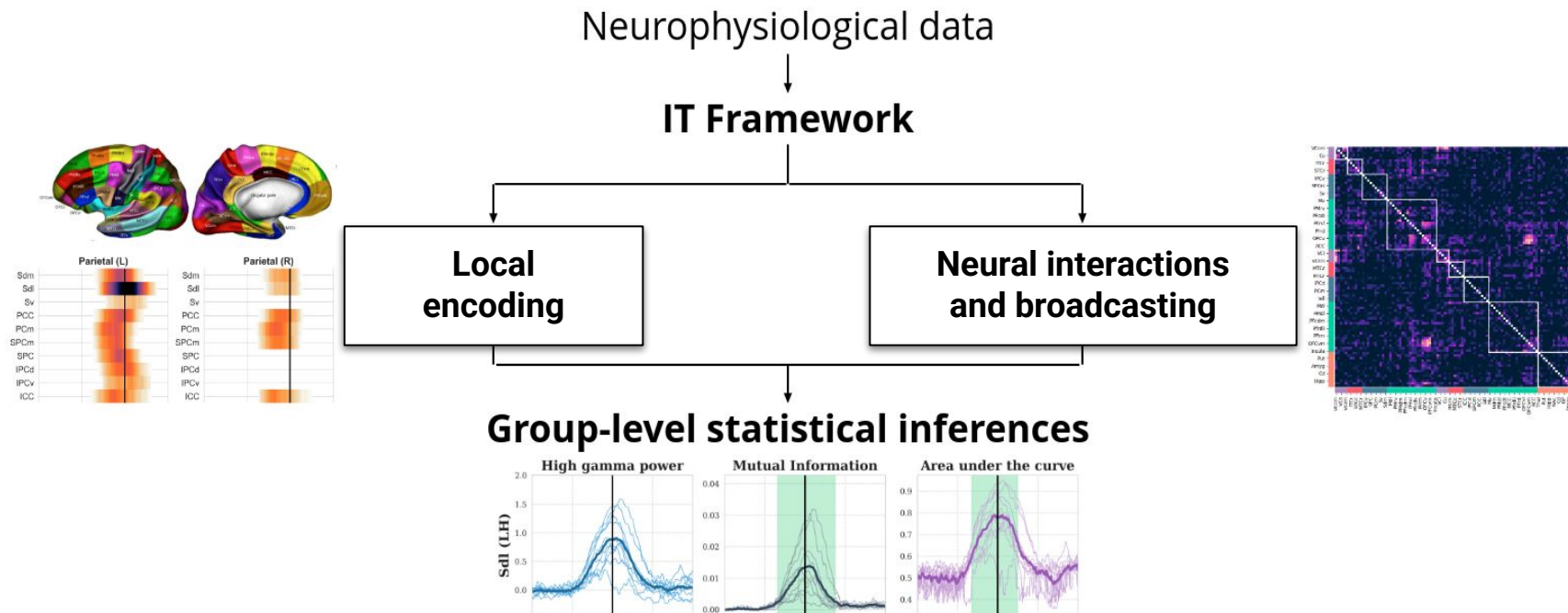
Reviewers:

- [@madvn](#)
- [@travisbthomp](#)

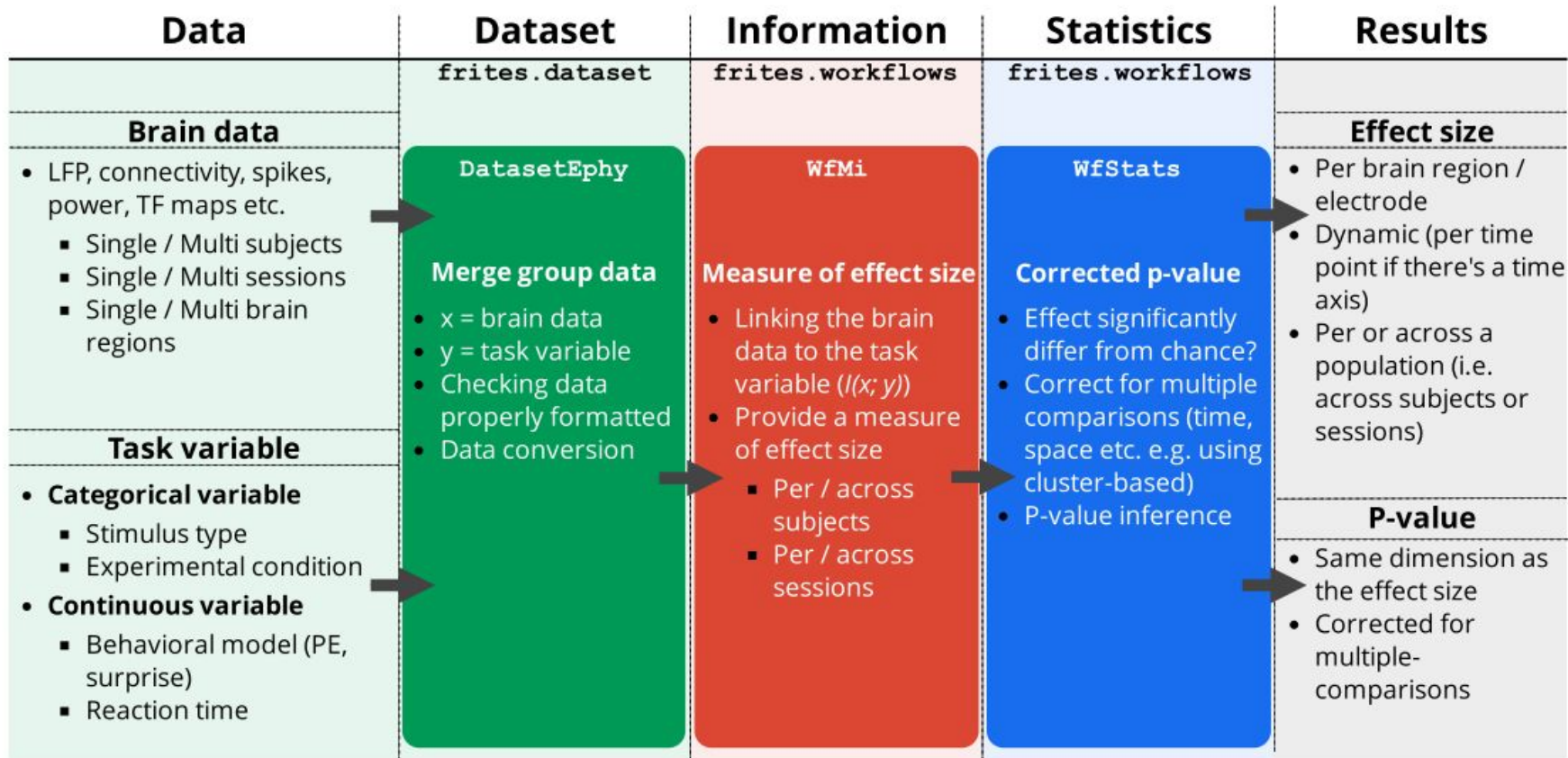
Summary

The field of cognitive computational neuroscience addresses open questions regarding the complex relation between cognitive functions and the dynamic coordination of neural activity over large-scale and hierarchical brain networks. State-of-the-art approaches involve the characterization of brain regions and inter-areal interactions that participate in cognitive processes ([Battaglia & Brovelli, 2020](#)). More precisely, the study of cognitive brain networks underlies linking local neural activity or interactions between brain regions to experimental variables, such as sensory stimuli or behavioral responses. The relation between the brain data and external variables might take complex forms (e.g. non-linear relationships) with strong variations across brain regions and participants. Therefore, powerful measures of information

Information theory for brain interactions



Typical Pipeline





Cooking Frites

<https://github.com/brainets/CookingFrites>

EBRAINS Academy Workshop Series

Session 1 – intracranial EEG

1. Login to the EBRAINS Collab

<https://wiki.ebrains.eu/>

2. Connect to EBRAINS Academy iEEG Collab and Drive

<https://wiki.ebrains.eu/bin/view/Collabs/ebrains-academy-workshop-ieeg/>

<https://wiki.ebrains.eu/bin/view/Collabs/ebrains-academy-workshop-ieeg/Drive>

3. Connect to EBRAINS lab

<https://lab.ebrains.eu/>

4. Connect to EBRAINS drive

<https://drive.ebrains.eu/>

Human Intracranial Database

release-5

Lachaux, J.; Rheims, S.; Chatard, B.; Dupin, M.; Bertrand, O.

Overview

[Data descriptor](#)[How to cite](#)[Get data](#)[Publications](#)[Specimen](#)DOI:  10.25493/FCPJ-NZ

License:

The use of this dataset requires that the user cites the associated DOI and adheres to the conditions of use that are contained in the Data Use Agreement. You may not use the dataset for commercial purposes.

Project: [Human Intracranial Database \(HID\)](#)Custodians:  [Lachaux, J.](#)

The Human Intracranial Database (HID) is a collection of stereotactic electroencephalography (sEEG) data in epileptic patients, performing up to eight behavioral tasks. The behavioral tasks were used as functional localizers: short and classic task paradigms designed to activate large-scale neural networks involved in language processing (LEC1 and LEC2), verbal and visuo-spatial working memory (MVEB and MVIS), visual attention (VISU), motor behavior (MOTO), high-level visual (MCSE) and auditory perception (AUDI). Furthermore, most patients were also recorded during resting state (REST) and a series of tests designed to evaluate whether specific types of actions can generate artifacts (ARFA). The sEEG data were obtained via several sEEG depth-electrodes (linear arrays with up to 20 contacts) that were implanted in each patient in a stereotactic surgery. Coordinates of the sEEG electrode contacts are provided in the individual brain space of each patient as well as in a reference brain space (MNI ICBM 152). Semantically sEEG electrode contacts were linked to brain areas of the MarsAtlas, but can be positioned on the individual anatomy for checking purposes.

Version specification:

This is the fifth release of the Human Intracranial Database. It is comprised of stereotactic electroencephalography (sEEG) data recorded in 100 epileptic patients performing behavioral tasks. The fifth release contains data from 20 new patients as compared to release-4. The data structure of this release is conform to the Brain Imaging Data Structure (BIDS) standard v1.2.0.

Study targets:

- [alpha activity](#)
- [beta activity](#)
- [gamma activity](#)
- [local field potential](#)
- [working memory](#)

[view all](#)

Behavioral protocols:

- [Auditory Perception Task \(Perrone-Bertolotti et al. 2012\)](#)
- [Visuo-Spatial Working Memory Task \(Hamamé et al. 2012\)](#)
- [Instructed Moto-Coordination Task \(Lachaux et al., 2020\)](#)
- [Resting State Paradigm \(Lachaux et al. 2020\)](#)
- [Sternberg's Verbal Working Memory Task \(Hamamé et al, 2012\)](#)

[view all](#)

Preparation: [in vivo](#)

Experimental approach:

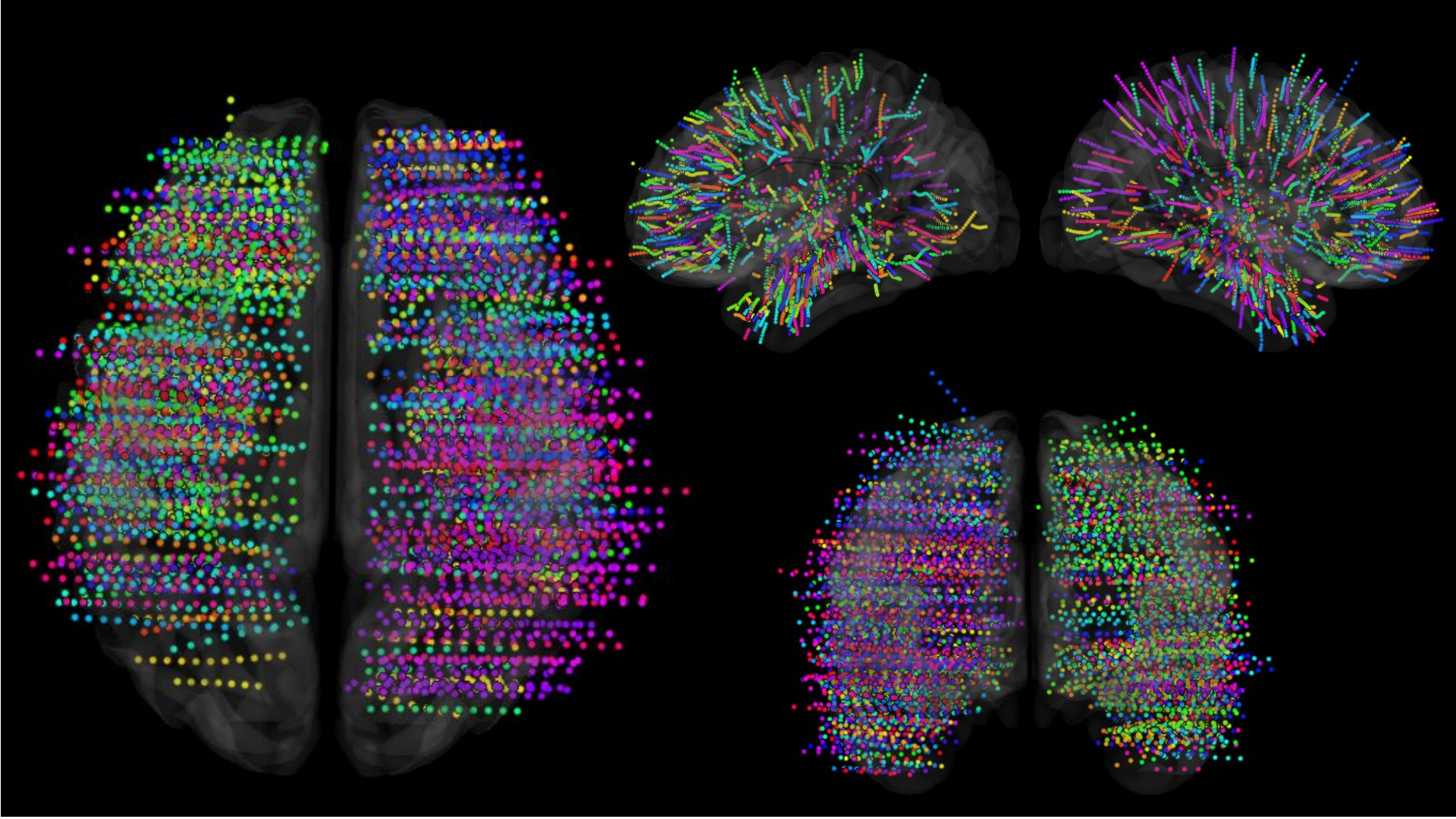
- [electrophysiology](#)
- [behavior](#)

Technique:

- [stereoelectroencephalography](#)
- [stereotactic surgery](#)
- [implant surgery](#)
- [magnetic resonance imaging](#)
- [transformation](#)

Keywords:

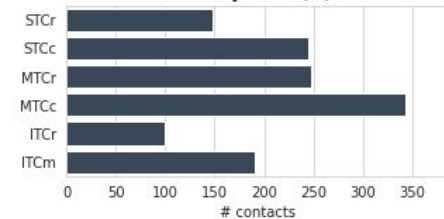
 [functional localizers](#)



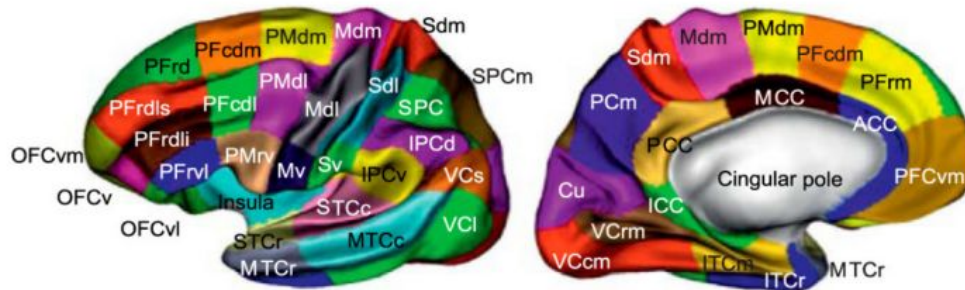
Frontal (L)



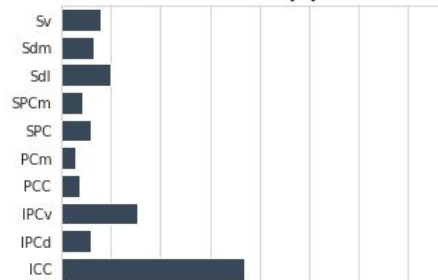
Temporal (L)



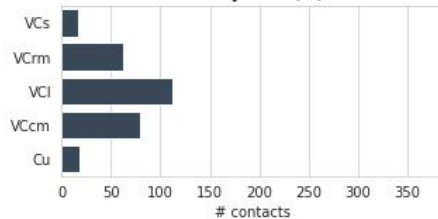
contacts across subjects



Parietal (L)



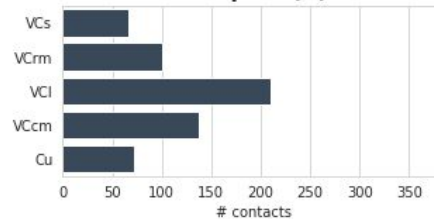
Occipital (L)



Parietal (R)



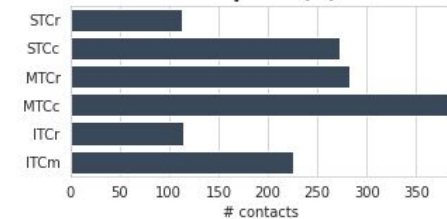
Occipital (R)

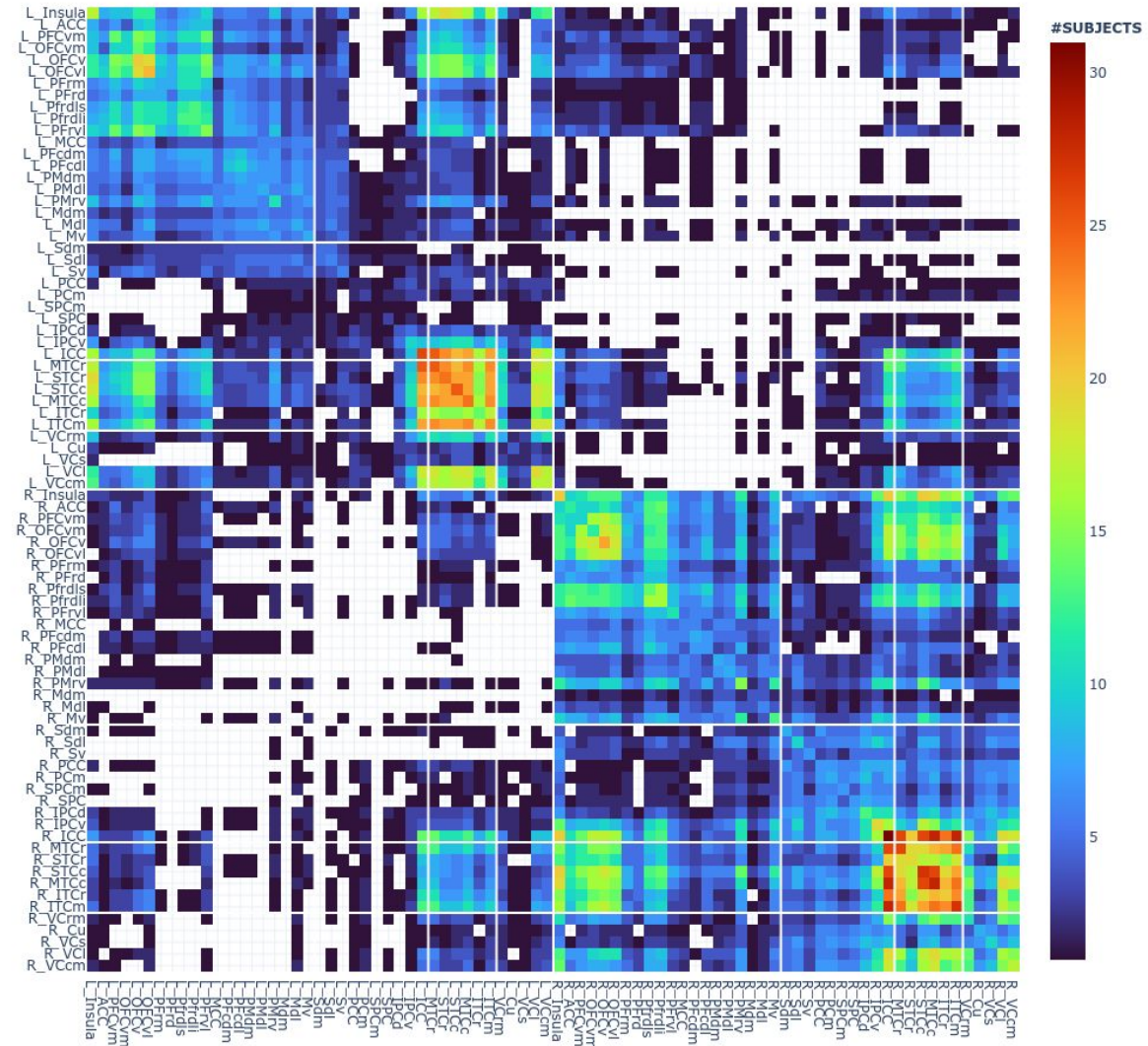


Frontal (R)



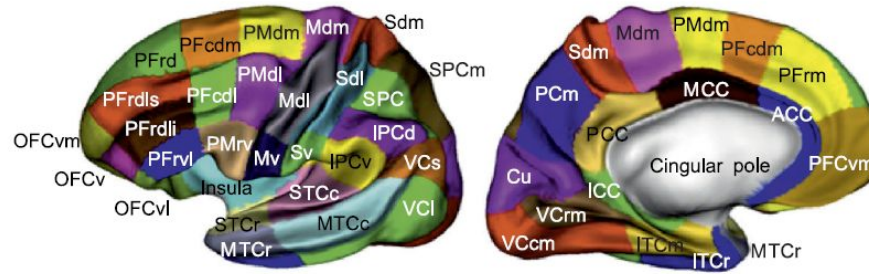
Temporal (R)



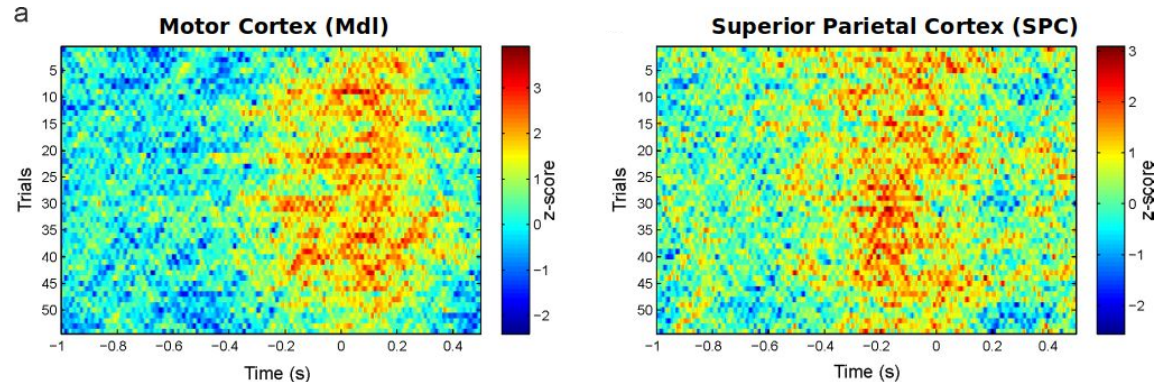
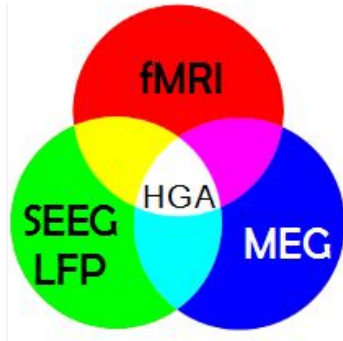


Single-trial and Atlas-based High-Gamma Activity

MarsAtlas



High-Gamma Activity (60-120Hz)



Visual Search

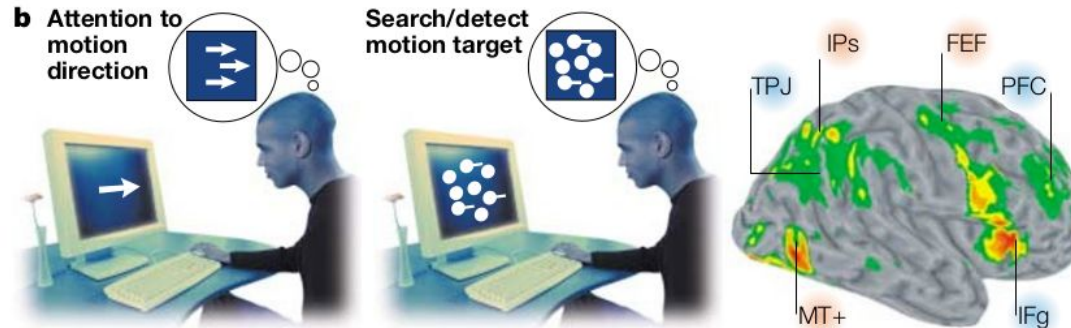
NATURE REVIEWS | NEUROSCIENCE

VOLUME 3 | MARCH 2002 | 201

CONTROL OF GOAL-DIRECTED AND STIMULUS-DRIVEN ATTENTION IN THE BRAIN

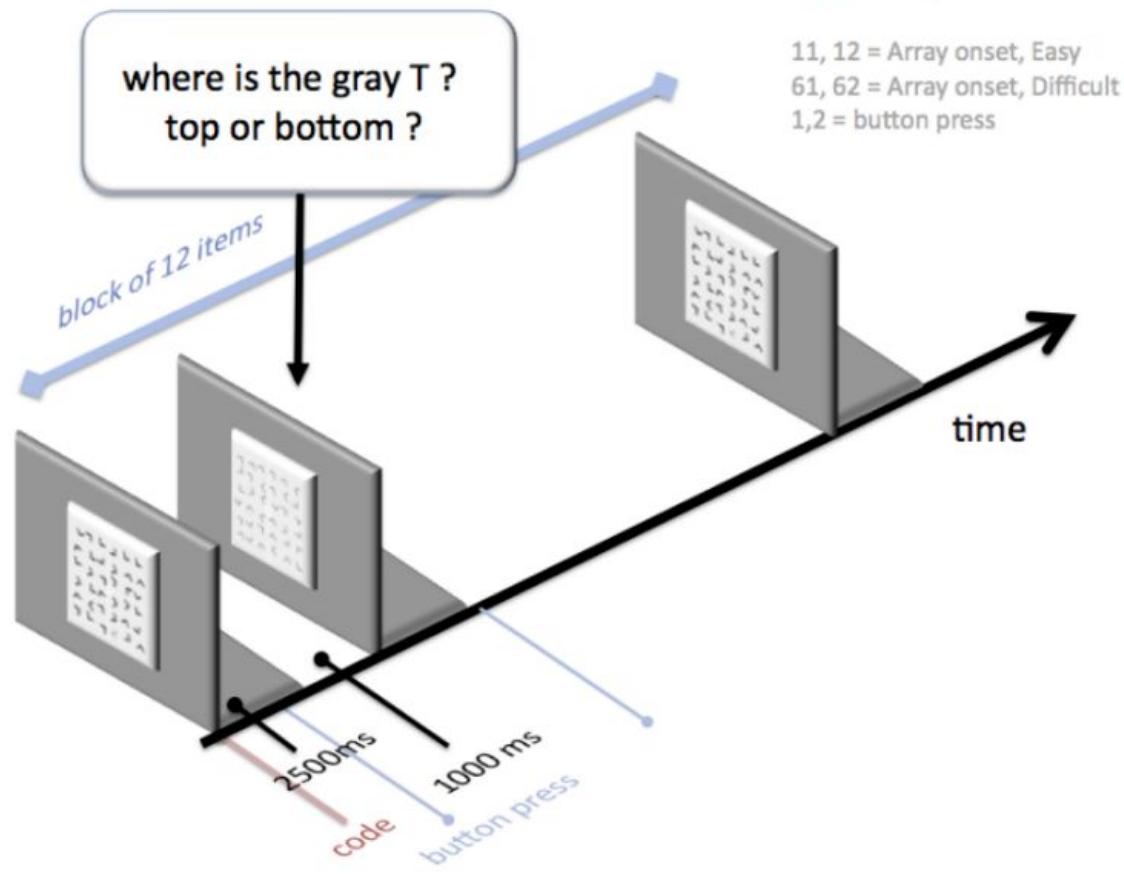
Maurizio Corbetta and Gordon L. Shulman

We review evidence for partially segregated networks of brain areas that carry out different attentional functions. One system, which includes parts of the intraparietal cortex and superior frontal cortex, is involved in preparing and applying goal-directed (top-down) selection for stimuli and responses. This system is also modulated by the detection of stimuli. The other system, which includes the temporoparietal cortex and inferior frontal cortex, and is largely lateralized to the right hemisphere, is not involved in top-down selection. Instead, this system is specialized for the detection of behaviourally relevant stimuli, particularly when they are salient or unexpected. This ventral frontoparietal network works as a 'circuit breaker' for the dorsal system, directing attention to salient events. Both attentional systems interact during normal vision, and both are disrupted in unilateral spatial neglect.



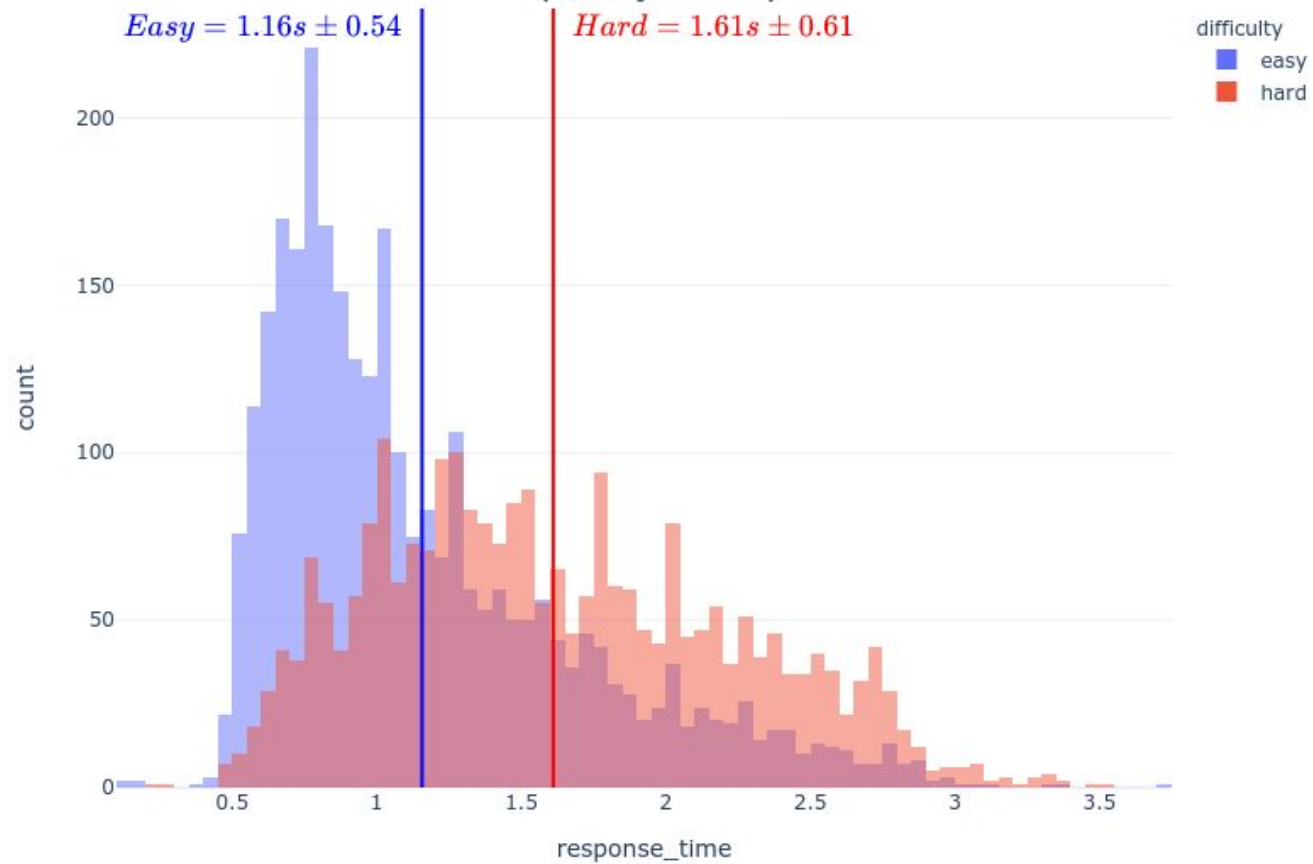
Human iEEG data - Visual search

MCSE

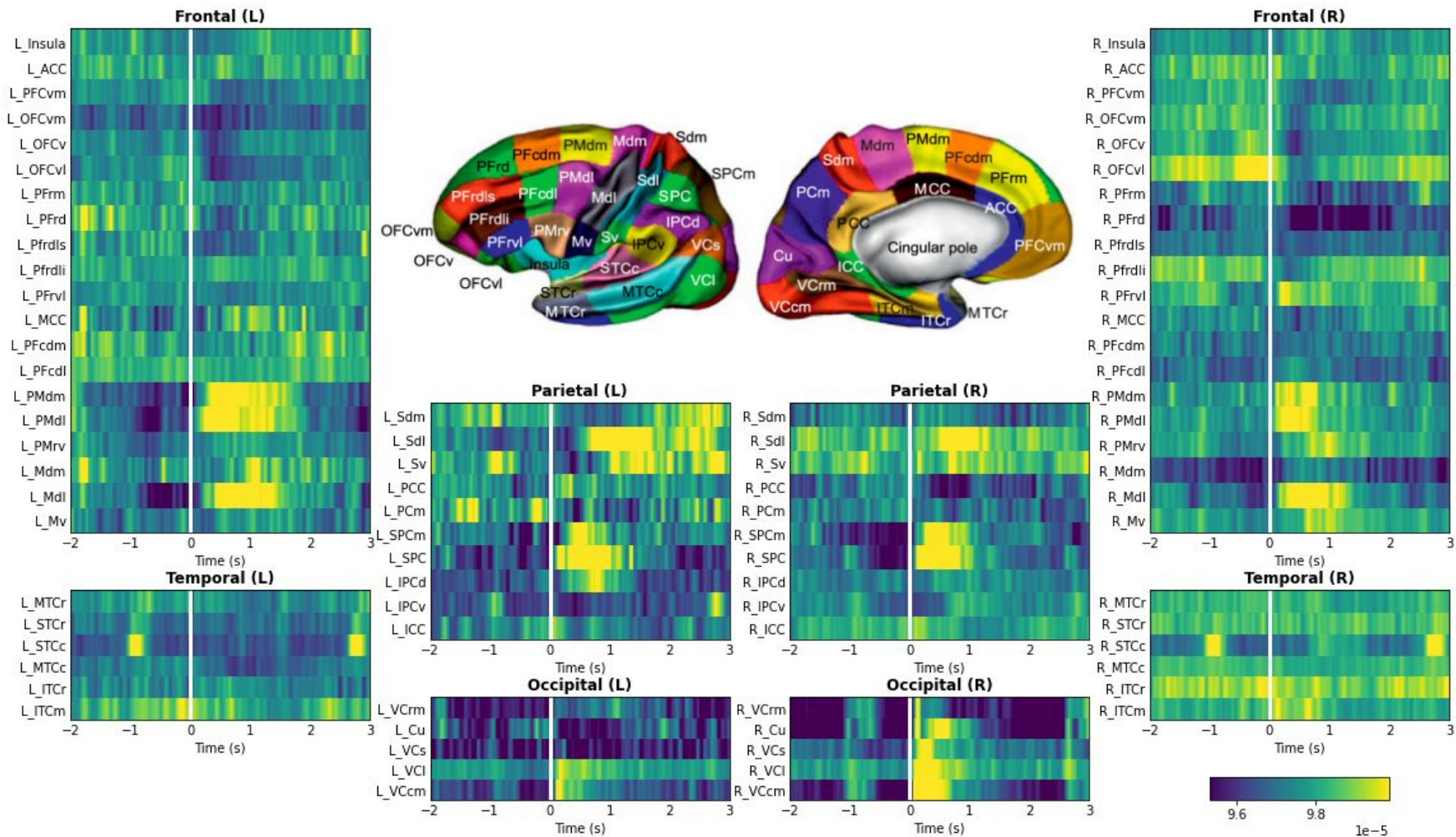


Single trial reaction_time over subjects

(n_subjects=65)



Power f50f150-sm0 aligned on stimulus presentation (#subject=62)



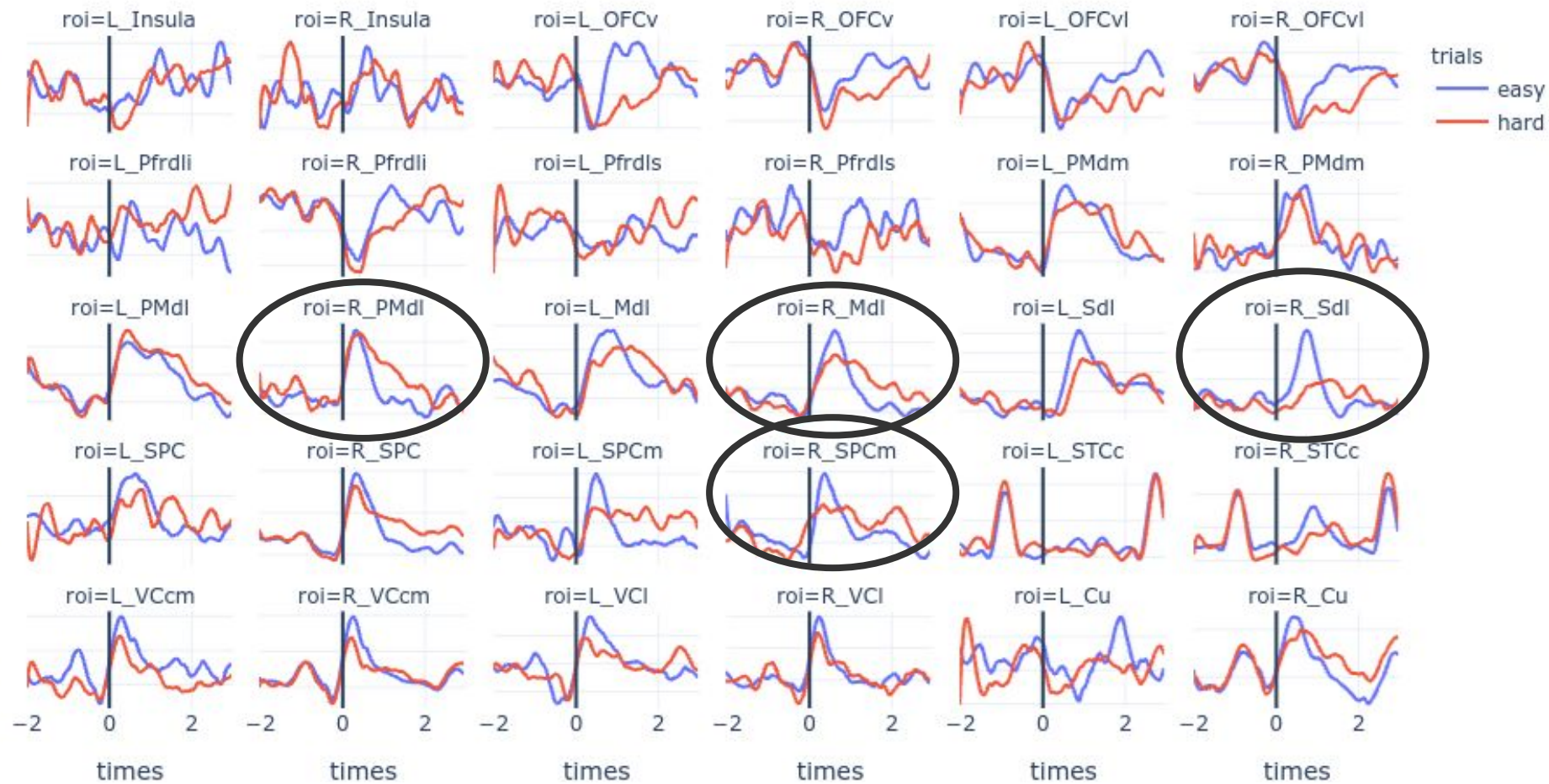
f50f150

f50f150

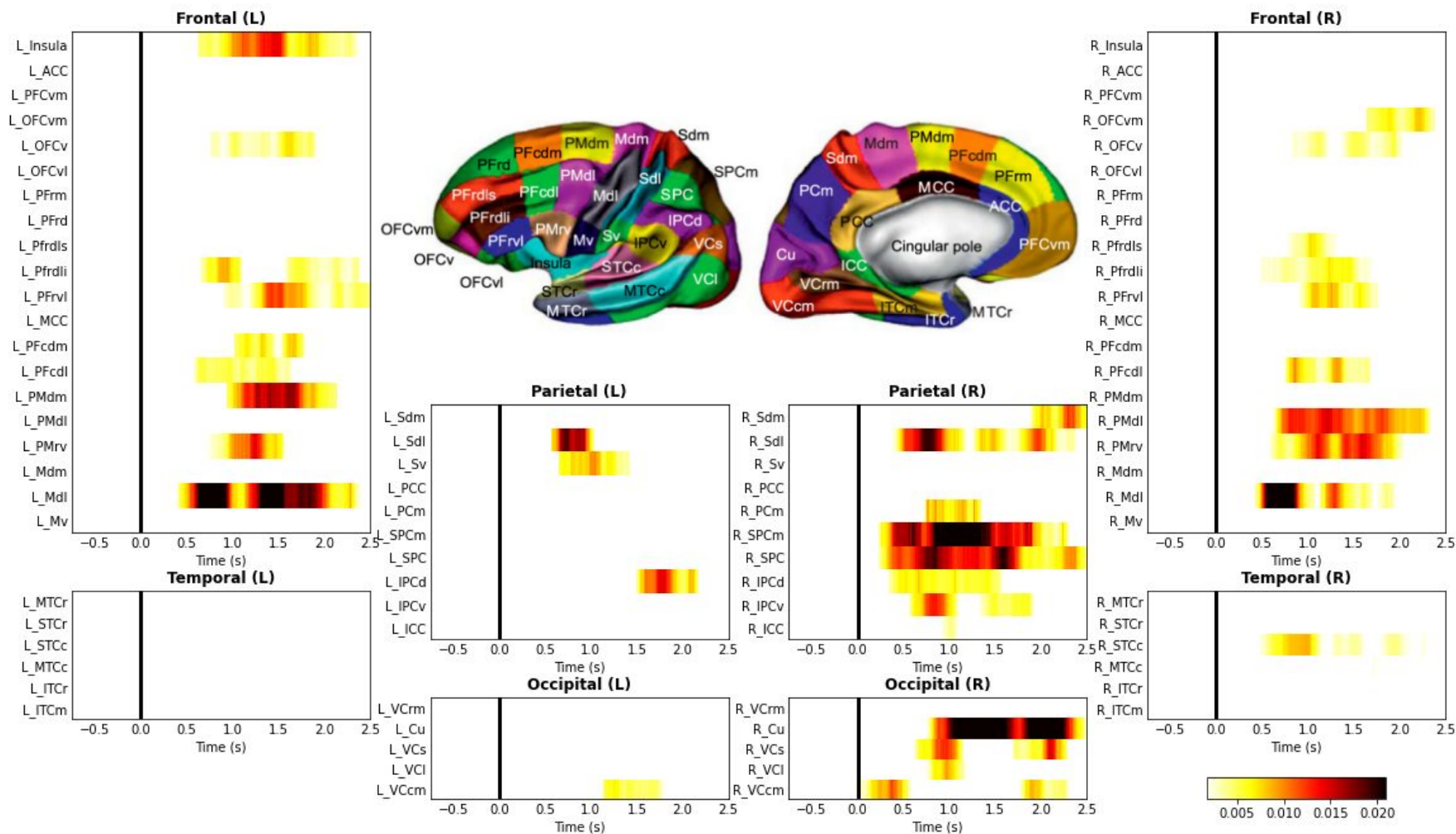
f50f150

f50f150

f50f150



I(gamma; Easy vs. Hard) - data aligned on stimulus presentation



Task-related Dynamic Functional Connectivity DCF (easy vs hard conditions)

I(DFC gamma; Easy vs. Hard) align on stimulus presentation (estimator=gcml, n_perm=200, mcp=cluster)

